

# Black Marble User Guide (Version 1.3)

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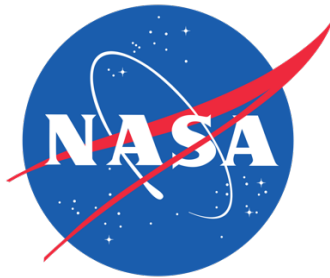
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## Acronyms

|         |                                                               |
|---------|---------------------------------------------------------------|
| AERONET | Aerosol Robotic Network                                       |
| AOD     | Aerosol Optical Depth                                         |
| BRDF    | Bidirectional Reflectance Distribution Function               |
| BRF     | Bidirectional Reflectance Factor                              |
| DMSP    | Defense Meteorological Satellite Program                      |
| DNB     | Day/Night Band                                                |
| EOS     | Earth Observing System                                        |
| GEO     | Group on Earth Observations                                   |
| GIS     | Geographic Information System                                 |
| HAM     | Half-angle Mirror                                             |
| HDF-EOS | Hierarchical Data Format - Earth Observing System             |
| IR      | Infrared                                                      |
| L1B     | Level-1 B                                                     |
| L2G     | Level-2 Gridded                                               |
| LAI     | Leaf Area Index                                               |
| LANCE   | The Land, Atmosphere Near real-time Capability for EOS        |
| LZA     | Lunar Zenith Angle                                            |
| JPSS    | Joint Polar Satellite System                                  |
| NASA    | National Aeronautics and Space Administration                 |
| NBAR    | Nadir BRDF-Adjusted Reflectance                               |
| NCEP    | National Centers for Environmental Prediction                 |
| NDVI    | Normalized Difference Vegetation Index                        |
| NDSI    | Normalized Difference Snow Index                              |
| NIR     | Near-infrared                                                 |
| NRT     | Near Real-Time                                                |
| NTL     | Nighttime Lights                                              |
| nW      | Nanowatt (One billionth ( $10^{-9}$ ) of a watt)              |
| Pgap    | Gap Fraction Probability                                      |
| PGE     | Product Generation Executable                                 |
| PRWGLP  | Puerto Rico's Working Group on Light Pollution                |
| QA      | Quality Assurance                                             |
| QF      | Quality Flag                                                  |
| RTA     | Rotating Telescope Assembly                                   |
| SDS     | Scientific Data Set                                           |
| SIPS    | Science Investigator-led Processing System                    |
| S-NPP   | Suomi National Polar-orbiting Platform                        |
| TOA     | Top of Atmosphere                                             |
| UTC     | Coordinated Universal Time                                    |
| VCM     | VIIRS Cloud Mask                                              |
| VCST    | VIIRS Calibration Support Team                                |
| VIIRS   | Visible Infrared Imaging Radiometer Suite                     |
| VNP09   | VIIRS Surface Reflectance product                             |
| VNP46   | NASA's Black Marble nighttime lights product suite            |
| VNP46A1 | Suomi-NPP Daily at-sensor TOA nighttime lights product        |
| VNP46A2 | Suomi-NPP Daily moonlight-adjusted nighttime lights product   |
| VNP46A3 | Suomi-NPP Monthly moonlight-adjusted nighttime lights product |
| VNP46A4 | Suomi-NPP Yearly moonlight-adjusted nighttime lights product  |

|         |                                                                    |
|---------|--------------------------------------------------------------------|
| VJ146A1 | NOAA-20/JPSS-1 Daily at-sensor TOA nighttime lights product        |
| VJ146A2 | NOAA-20/JPSS-1 Daily moonlight-adjusted nighttime lights product   |
| VJ146A3 | NOAA-20/JPSS-1 Monthly moonlight-adjusted nighttime lights product |
| VJ146A4 | NOAA-20/JPSS-1 Yearly moonlight-adjusted nighttime lights product  |

## 1 Introduction

The Day/Night Band (DNB) sensor of the Visible Infrared Imaging Radiometer Suite (VIIRS), on-board the Suomi-National Polar-orbiting Partnership (S-NPP) and Joint Polar Satellite System (JPSS) satellite platforms, provides global daily measurements of nocturnal visible and near-infrared (NIR) light that are suitable for Earth system science and applications studies. The VIIRS DNB's ultra-sensitivity in lowlight conditions allows for the generation of new science-quality nighttime products as a result of significant improvements to sensor resolution and calibration compared to those provided previously by the Defense Meteorological Satellite Program's (DMSP) generation of nighttime lights imagery. These improvements allow us to better monitor both the magnitude and signature of nighttime phenomena and anthropogenic sources of light emissions.

Since the launch of the S-NPP satellite in 2011, multiple studies have used the VIIRS DNB as the primary data source in a wide range of study topics. They include: (1) feature extraction techniques to detect severe weather impacts to urban infrastructure (Cao et al., 2013; Cole et al., 2017; Mann et al., 2016; Molthan and Jedlovec, 2013); (2) detection of sub-pixel scale features, e.g., fires (Polivka et al., 2016), shipping vessels (Asanuma et al., 2016; Elvidge et al., 2015; Straka et al., 2015), lightning flashes (Bankert et al., 2011), surface oil slicks (Hu et al., 2015), and gas flares (Elvidge et al., 2015; Liu et al., 2017, Liu et al., 2017); and (3) techniques for monitoring nighttime atmospheric optical properties, including clouds (Minnis et al., 2016; Walther et al., 2013), aerosols (Johnson et al., 2013; McHardy et al., 2015), particulate matter (Wang et al., 2016), and gravity waves in the upper atmosphere via nightglow (Miller et al., 2015).

As with early research that utilized the DMSP 's Operational Line Scanner (OLS) (Huang et al., 2014), recent studies using the VIIRS DNB have employed statistical analyses and correlation discovery methods to confirm established empirical relationships with a wide range of human-linked patterns and processes. They include socioeconomic variables (Chen and Nordhaus, 2015; Chen et al., 2015; Levin and Zhang, 2017; Li et al., 2013; Ma et al., 2014; Shi et al., 2014; Yu et al., 2015), as well as changes driven by urban expansion (Guo et al., 2015; Sharma et al., 2016; Shi et al., 2014), energy use (Coscieme et al., 2014; Román and Stokes, 2015), and carbon emissions (Oda et al., 2017; Ou et al., 2015).

To realize the full potential of the VIIRS DNB time series record, NASA has developed a new suite of standard products that represent the current state-of-the-art in nighttime lights (NTL) applications, NASA's Black Marble product suite (VNP46/VJ146). NASA's Black Marble nighttime lights product, at 15 arc-second spatial resolution, is available from January 2012-present with data from the VIIRS DNB

sensor. The VNP46/VJ146 product suite is being processed on a daily basis within 3-5 hours of acquisition, which enables both near-real-time uses and long-term monitoring applications. The VNP46/VJ146 product suite includes the daily at-sensor top of atmosphere (TOA) nighttime lights (NTL) product (VNP46A1/VJ146A1), daily moonlight-adjusted nighttime lights product (VNP46A2/VJ146A2), monthly moonlight-adjusted nighttime lights product (VNP46A3/VJ146A3), and yearly moonlight-adjusted nighttime lights product (VNP46A4/VJ146A4). The retrieval algorithm, developed and implemented for routine global processing at NASA's Land Science Investigator-led Processing System (SIPS), utilizes all high-quality, cloud-free, atmospheric-, terrain-, vegetation-, snow-, lunar-, and stray light-corrected radiance to estimate daily nighttime lights and other intrinsic surface optical properties. The VIIRS Black Marble product has been used for global mapping of human activity patterns, such as tracking shipping and fishing vessels, gas flares, in addition to their application to humanitarian efforts, such as assessment of conflict-associated demographic changes and mapping impoverishment.

This user guide provides an overview of NASA's new VIIRS Level 3 Black Marble nighttime lights product suite (VNP46) to users. This document describes the theoretical basis for the algorithms, the operational processing, evaluation and validation of the product, and how to access the product. Additional details are available in related publications in section 10.

## 2 Algorithm

### 2.1 Overview of the Algorithm

NASA's operational Black Marble product suite ingests multiple-source input datasets and ancillary data to output the highest quality pixel-based estimates of NTL. These NTL estimates are accompanied by pixel-level quality flags. The principal features of the algorithm are summarized in the following sections. More details of the algorithm are available in related publications in section 10.

NASA's Black Marble algorithm produces cloud-free images that have been corrected for atmospheric, terrain, lunar BRDF, thermal, and straylight effects. The corrected nighttime radiance, resulting in a superior retrieval of nighttime lights at short time scales and a reduction in background noise, enables quantitative detection and analyses of daily, seasonal and annual variations. Key algorithm enhancements include: (1) lunar irradiance modeling to resolve non-linear changes in phase and libration; (2) vector radiative transfer and lunar bidirectional surface anisotropic reflectance modeling to correct for atmospheric and bidirectional reflectance distribution function (BRDF) effects; (3) geometric-optical and canopy radiative transfer modeling to account for seasonal variations in NTL; and (4) temporal gap-filling to reduce persistent data gaps.



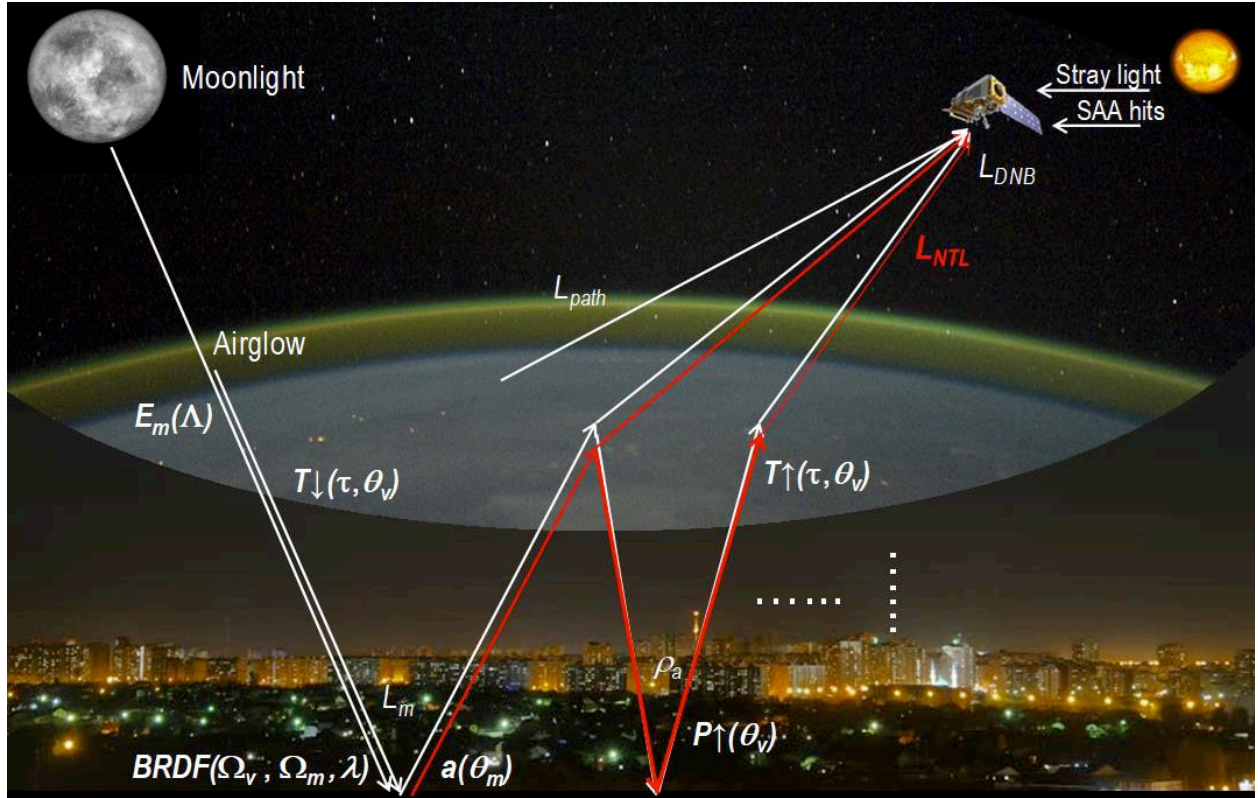
## 2.2 Atmospheric Correction

NASA’s Black Marble retrieval strategy combines daytime VIIRS DNB surface reflectance, BRDF, surface albedo, nadir BRDF-adjusted reflectance (NBAR), and lunar irradiance values to minimize the biases caused by extraneous artifacts in the VIIRS NTL time series record.

Using this novel “turning off the Moon” approach, illustrated in Figure 1, the upward surface radiance from artificial light emissions,  $L_{NTL}$  [units of  $\text{nWatts} \cdot \text{cm}^{-2} \cdot \text{sr}^{-1}$ ], can be extracted from at-sensor nighttime radiance at TOA,  $L_{DNB}$ , using the following equation:

$$L_{NTL} = \left[ \left( \frac{L_{DNB} - L_{path}}{T_{\uparrow}(\tau, \theta_v)} \right) (1 - a(\theta_m) \rho_a) - L_m T_{\downarrow}(\tau, \theta_v) \right] / P_{\uparrow}(\theta_v) \quad (1)$$

where  $L_{path}$  is the nighttime path radiance (*i.e.*, the radiance generated by scattering within the atmosphere), and  $a(\theta_m)$  is the VIIRS-derived actual (or blue-sky) surface albedo; incorporating the directional influence of sky radiance and multiple scattering effects between the ground and the atmosphere (Román et al., 2010). For the latter, a snow albedo retrieval scheme is used if the VIIRS current day snow status flag is activated (Klein and Stroeve, 2002; Liu et al., 2017, Liu et al., 2017; Moustafa et al., 2017; Wang et al., 2012).  $P_{\uparrow}(\theta_v)$  is the probability of the upward transmission of NTL emissions through the urban vegetation canopy, defined in Equation 4. The atmospheric backscatter is given by  $\rho_a$ , and  $T_{\downarrow}(\tau, \theta_v)$  and  $T_{\uparrow}(\tau, \theta_v)$  are the total transmittance (including direct and diffuse radiation) along the lunar-ground and ground-sensor paths, respectively. The latter two are a function of view-illumination geometry and the total atmospheric column optical depth ( $\tau$ ) due to mixed gases, water vapor, and aerosol particles. The retrieval uses a modified algorithm based on the VIIRS Surface Reflectance product (VNP09) to estimate the values of  $L_{path}$ ,  $\rho_a$ ,  $T_{\downarrow}(\tau, \theta_v)$ , and  $T_{\uparrow}(\tau, \theta_v)$  for a given set of surface and atmospheric conditions (Roger et al., 2016; Skakun et al., 2018). Additional input datasets include the standard VIIRS Cloud Mask (VCM) (Kopp et al., 2014), atmospheric profiles obtained from the National Centers for Environmental Prediction (NCEP) model (*i.e.*, water vapor, ozone, and surface pressure) (Moorthi et al., 2001), and the VIIRS aerosol model combined with daytime-to-daytime averaged aerosol optical depth (AOD) at  $0.550 \mu\text{m}$  to extrapolate the nighttime AOD.



**Figure 1** Overview of NASA's Black Marble retrieval strategy (*cf.*, Equation 1). During the ~50% portion of the lunar cycle when moonlight is present at the time of satellite observation, the surface upward radiance from artificial light emissions,  $L_{NTL}$  [units of  $\text{nWatts} \cdot \text{cm}^{-2} \cdot \text{sr}^{-1}$ ], can be extracted from at-sensor nighttime radiance at TOA ( $L_{DNB}$ ).  $L_{path}$  is the nighttime path radiance,  $a(\theta_m)$  is the VIIRS-derived actual surface albedo. The atmospheric backscatter is given by  $\rho_a$ .  $T_{\downarrow}(\tau, \theta_v)$  and  $T_{\uparrow}(\tau, \theta_v)$  are the total transmittances along the lunar-ground and ground-sensor paths (respectively).  $P_{\uparrow}(\theta_v)$  is the probability of the upward transmission of NTL emissions through the urban vegetation canopy.

## 2.3 BRDF Correction

The Black Marble algorithm estimates the actual moonlight, aerosol, and surface albedo contribution through analytical BRDF model inversion. This model has proven effective in removing biases introduced by extraneous sources of nighttime light emissions.

The surface BRDF, or reflectance anisotropy is governed by the angle and intensity of illumination – whether that illumination is solar or lunar or from airglow – and by the structural complexity of the surface, resulting in variations in brightly illuminated regions and highly shadowed areas. The semi-empirical RossThick-LiSparse Reciprocal (RTLSR, or Ross-Li) BRDF model (Román et al., 2010; Roujean et al., 1992; Schaaf et al., 2002, Schaaf et al., 2011; Strahler et al., 1999) is advantageous in this regard since (1) it is the most likely kernel-driven combination to capture the wide range of conditions affecting the VIIRS DNB on a global basis; (2) it allows robust analytical model inversion

with a pixel-specific estimate of the uncertainty in the model parameters and linear combinations thereof (Lucht and Roujean, 2000); and (3) the scheme is flexible enough to allow us to easily adapt other kernels should any become available, and demonstrate its superiority for a particular scenario.

For VIIRS DNB acquisitions over snow-free and snow-covered surfaces, we define the spectral radiance contribution from moonlight,  $L_m$ ,

$$L_m(\Omega_v, \Omega_m, \Lambda) = \frac{E_m(\Lambda)}{\pi} \mathbf{BRF}(\Omega_v, \Omega_m, \Lambda) \cos(\Theta_m) \quad (2)$$

in terms of the Ross-Li model:

$$\mathbf{BRDF}(\Omega_v, \Omega_m, \Lambda) \approx \frac{\mathbf{BRF}(\Omega_v, \Omega_m, \Lambda)}{\pi} = f_{iso}(\Lambda) + f_{vol}(\Lambda) K_{vol}(\Omega_v, \Omega_m) + f_{geo}(\Lambda) K_{geo}(\Omega_v, \Omega_m) \quad (3)$$

Here, we define the wavelength for the narrowband instrument of interest as the weighted center,  $\Lambda$ , of the VIIRS DNB spectral band [0.5–0.9  $\mu\text{m}$ ]. Parameter  $f_{iso}(\Lambda)$  is the isotropic scattering component and equal to the bidirectional reflectance for a pixel viewing zenith angle  $\theta_v = 0$  and a lunar zenith angle  $\theta_m = 0$ . Parameter  $f_{geo}(\Lambda)$  is the coefficient of the LiSparse-reciprocal geometric scattering kernel  $K_{geo}$ , derived for a sparse ensemble of surface casting shadows on a Lambertian background (Li and Strahler, 1992). Parameter  $f_{vol}(\Lambda)$  is the coefficient for the RossThick volume scattering kernel  $K_{vol}$ , so-called for its assumption of a dense leaf canopy (Ross, 2012).

To achieve a high-quality BRDF retrieval, NASA's Black Marble algorithm collects all available daytime, atmospherically-corrected, VIIRS DNB bidirectional reflectance factor (BRF) over a multi-date period (normally 16-days) to establish the analytical solution for the Ross-Li BRDF model parameter values,  $f_k(\Lambda)$ . Note that during moon-free nights, when atmospheric airglow is the dominant emission source, the VNP46 algorithm sets the illumination geometry to near-nadir ( $\theta_m = 10^\circ$ ) and the lunar irradiance to  $E_m(\Lambda) = 0.26 \text{ nW} \cdot \text{m}^{-2}$  (Liao et al., 2013). This enables a BRDF correction even in the absence of moonlight.

### 2.3 Seasonal Vegetation Correction

Another known source of uncertainty in retrieving satellite-derived NTL is the influence of canopy-level foliage within the ground-to-sensor geometry path (Román and Stokes, 2015). This effect, which has been shown to reduce the magnitude of NTL at city-wide scales (Levin, 2017; Levin and Zhang, 2017), is most pronounced in temperate urban regions, where mixed and deciduous vegetation are most pervasive. Given its seasonal dependence, this occlusion effect (obscuration of surface light by foliage) should be proportional in magnitude to the density and vertical distribution pattern of leaves within a given VIIRS

DNB pixel. Hence, while the effect may be non-linear (due to the confluence of factors that control the seasonality, physiognomy, and vertical distribution of urban vegetation canopies), the effect can be parameterized using analytical models, which aim to retrieve canopy structure parameters from multi-angle remote sensing data (Chopping, 2006). With this concept in mind, we are employing a vegetation dispersion parameter, known as the clumping index,  $\psi$ , to parameterize the confined distribution of foliage within distinct canopy structures (Chen et al., 2005; Chen and Black, 1991; Jiao et al., 2018; Leblanc et al., 2005; Nilson, 1971):

$$P_{\uparrow}(\theta_v) = e^{-\psi G(\theta_v) LAI} / \cos(\theta_v) \quad (4)$$

Here,  $P_{\uparrow}(\theta_v)$  is the probability of the upward transmission of NTL emissions through the urban vegetation canopy (known as the gap fraction probability and hereafter termed the  $P_{gap}$  equation),  $G(\theta_v)$  is the extinction coefficient that expresses the mean area projection of plant elements in the direction  $\theta_v$  (being 0.5 for canopies with a random distribution of leaf angles), and  $LAI$  is the Leaf Area Index.

The  $P_{gap}$  equation can be inverted from available daily VIIRS BRDF-derived clumping index values, as done in Hill et al. (2011) and He et al. (2012). The VIIRS LAI retrievals are based on the current standard product (VNP15) (Park et al., 2017). In case of poor-quality or missing LAI values (*e.g.*, when LAI is not retrieved over dense urban areas), we are employing the VIIRS LAI backup algorithm by using a look-up table (LUT) (Knyazikhin et al., 1999; Xiao et al., 2016) with normalized difference vegetation index (NDVI) generated from high-quality retrievals from the VIIRS NBAR product (Shuai et al., 2013).

## 2.4 Monthly and Yearly Nighttime Light Composite

Monthly and yearly NTL composites (Wang et al., 2022) are generated from daily atmospherically- and lunar-BRDF-corrected NTL radiance to remove the influence of extraneous artifacts and biases. NTL outliers are excluded according to Boxplot metrics (Tukey 1977). The observations that fall out of the range of  $Q1 - 1.5 \cdot IQR$  and  $Q3 + 1.5 \cdot IQR$  are identified as outliers and excluded from the NTL composite. Interquartile range (IQR) score is the range between 25th (Q1) and the 75th (Q3) percentile. Unlike Z-score (Kreyszig 1979), the IQR score method does not require normal distribution of the observations. The monthly and yearly NTL composite are then calculated from the mean values of the remaining observations. To remove any residual background noise, the NTL composite values with radiances less than  $0.5 \text{ nW} \cdot \text{cm}^{-2} \cdot \text{sr}^{-1}$  are set to zero. Aurora-contaminated pixels are filled with gap-filled values.

User's should be aware that artificial lights derived from VIIRS DNB data show a strong angular effect (Li et al., 2019; Wang et al., 2021), impacting retrievals, particularly across dense urban centers where NTL radiance at nadir can be significantly higher than off-nadir observations. The presence of nighttime

snow also enhances the scattering of reflected NTL due to the increased surface reflectance. Accordingly, NASA's Black Marble monthly and yearly NTL composites are generated for multiple view-angle categories (i.e., near-nadir, off-nadir, and all angles) and snow status (snow-covered and snow-free) along with ancillary metrics of standard deviation, the number of observations, and mandatory QA flags. Users are encouraged to carefully use these NTL composite values (either separately or jointly) to meet their specific science research and application needs.

### **3 Data Product Formats**

NASA's Black Marble product suite includes the daily at-sensor TOA nighttime radiance (VNP46A1/VJ146A1), the daily moonlight and atmosphere-corrected NTL (VNP46A2/VJ146A2), monthly moonlight and atmosphere-corrected NTL (VNP46A3/VJ146A3), and yearly moonlight and atmosphere-corrected NTL (VNP46A4/VJ146A4) products at a 15 arc-second geographic linear latitude/longitude (lat/lon) grid. The data are provided in the standard land Hierarchical Data Format - Earth Observing System (HDF-EOS) format.

#### **3.1 Metadata**

Metadata (data attributes) provide information about data acquisition, input products, geographic location, the output of the data product, satellite instrument, processing environment, and other aspects of the retrieval. More details of the VNP46A1/VJ146A1, VNP46A2/VJ146A2, VNP46A3/VJ146A3, and VNP46A4/VJ146A4 product metadata are listed in Appendix A, B, C, and D.

#### **3.2 Filenames**

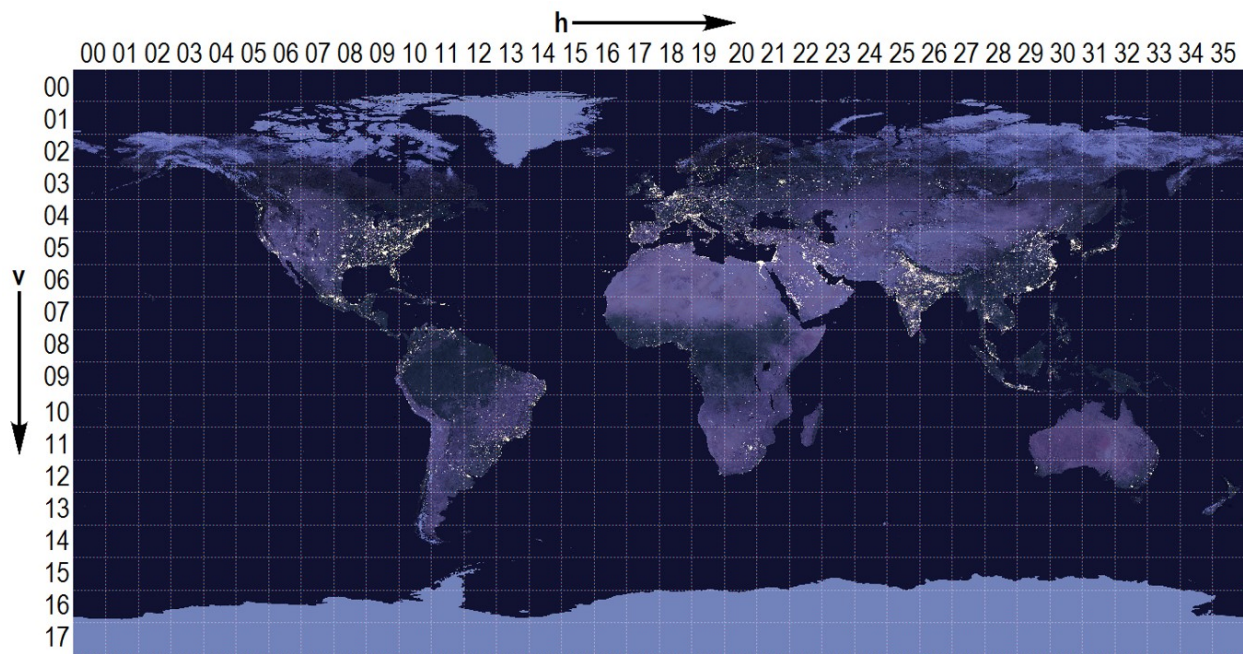
The filenames follow a naming convention, which gives useful information regarding the specific product. For example, the filename VNP46A1.A2015001.h08v05.001.2017012234657.h5 indicates:

- (1) VNP46A1: Product Short Name;
- (2) A2015001: Julian Date of Acquisition (A-YYYYDDD);
- (3) h08v05: Tile Identifier (horizontalXXverticalYY);
- (4) .001: Collection Version;
- (5) .2017012234657: Julian Date of Production (YYYYDDHHMMSS);
- (6) .h5: Data Format (HDF5).



### 3.3 Projection

NASA's Black Marble product suite employs the standard VIIRS science algorithms and software that produce the DNB standard (radiance-based) products, and their corresponding ancillary layers in gridded (Level 2G, Level 3) linear lat/lon format (Figure 2). The gridding algorithms were modified to work with the VIIRS DNB's unique viewing geometry, which, unlike the VIIRS moderate and imagery bands, has a ground pixel footprint at a nearly constant size (742 m). The rationale behind the VIIRS DNB gridding approach is to select the nighttime observations from available 6-min swath granules (2366 km along track, ~3100 km across-track), that are of high quality as indicated by the quality flags, and are the least affected by cloud cover and off-nadir viewing observations. The goal is to increase signal-to-noise, while maximizing coverage within a cell of the gridded projection (Tan et al., 2006; Wolfe et al., 2002). By implementing this combined gridding strategy and geographic linear lat/lon projection formats, we seek to improve the efficiency of processing and reprocessing the VNP46 product suite, preserve the satellite location and observation footprints, while also enabling the ingest of the products into accessible software for geographic information system (GIS)-friendly analysis and mapping.



**Figure 2** The Suomi-NPP VIIRS linear latitude/longitude (or geographic) grid consists of 460 non-overlapping Land tiles, which measure approximately  $10^{\circ} \times 10^{\circ}$  region.

### 4. Product generation

Data product inputs to NASA's Black Marble algorithm are listed in Table 1 and Table 2. The algorithm processing flow is depicted in Figure 3. The algorithm processing cycle is divided into daytime and

nighttime branches, and each processing branch produces a unique set of ancillary and quality assurance (QA) flags.

**Table 1** Black Marble VNP46A1 product input files

| <b>Input File</b>                   | <b>Description</b>                                                      |
|-------------------------------------|-------------------------------------------------------------------------|
| VNP02DNB                            | VIIRS/NPP Day/Night Band 6-Min L1B Swath 750m (L1B DNB)                 |
| VNP02MOD                            | VIIRS/NPP Moderate Resolution 6-Min L1B Swath 750m (L1B moderate bands) |
| NPP DNB <sup>*</sup>                | VIIRS L2G DNB radiance                                                  |
| NPP DNB angles <sup>*</sup>         | VIIRS L2G DNB angles                                                    |
| NPP MOD <sup>*</sup>                | VIIRS L2G moderate bands M10, M11, M12, M13, M15, M16                   |
| NPP PTDN <sup>*</sup>               | VIIRS DNB pointer files                                                 |
| The standard VIIRS Cloud Mask (VCM) | VIIRS cloud mask                                                        |

<sup>\*</sup>IP products.

**Table 2** Black Marble VNP46A2 product input files

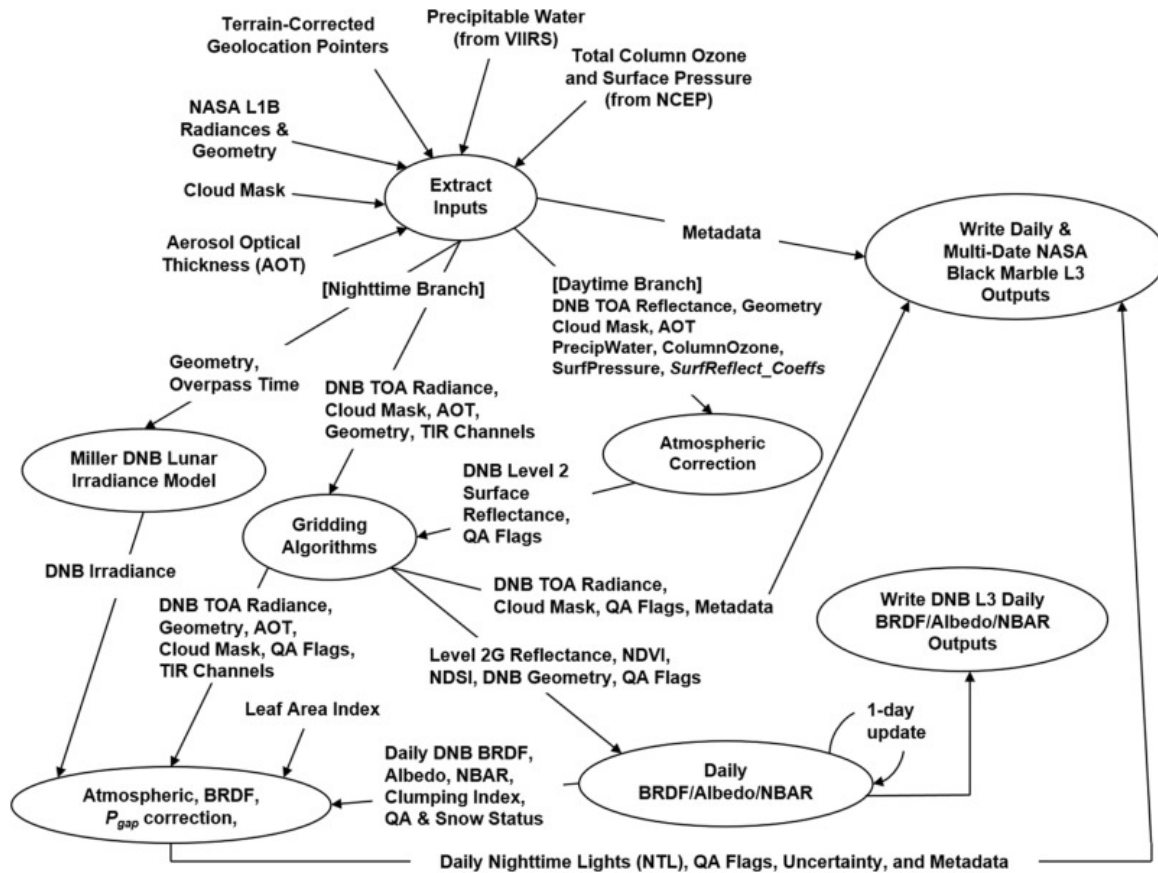
| <b>Input file</b> | <b>Description</b>                                                        |
|-------------------|---------------------------------------------------------------------------|
| VNP46A1           | VIIRS/NPP TOA Daily Gridded Day Night Band Linear Lat Lon Grid Night      |
| VNP43LGD NBA1     | VIIRS/NPP DNB BRDF/Albedo Model Parameters Daily L3 Global LLL Grid       |
| VNPLG09GA         | VIIRS/NPP Surface Reflectance Daily L2G Global Linear Lat Lon Grid        |
| VNP04LGA          | VIIRS/NPP Aerosols Optical Thickness Daily L2G Global Linear Lat Lon Grid |

**Table 3** Black Marble VNP46A3 product input files

| <b>Input file</b> | <b>Description</b>                                                                                  |
|-------------------|-----------------------------------------------------------------------------------------------------|
| VNP46A1           | VIIRS/NPP TOA Daily Gridded Day Night Band Linear Lat Lon Grid Night                                |
| VNP46A2           | VIIRS/NPP Moonlight and Atmosphere corrected Daily Gridded Day Night Band Linear Lat Lon Grid Night |

**Table 4** Black Marble VNP46A4 product input files

| <b>Input file</b> | <b>Description</b>                                                                                  |
|-------------------|-----------------------------------------------------------------------------------------------------|
| VNP46A1           | VIIRS/NPP TOA Daily Gridded Day Night Band Linear Lat Lon Grid Night                                |
| VNP46A2           | VIIRS/NPP Moonlight and Atmosphere corrected Daily Gridded Day Night Band Linear Lat Lon Grid Night |



**Figure 3** Algorithm processing cycle and ancillary parameters used by NASA's Black Marble product suite.

For the daytime branch, science product generation executables (PGEs) based on the standard suite of VIIRS land products are integrated as part of NASA's Black Marble processing chain. First, a modified version of the operational VIIRS surface reflectance algorithm (Roger et al., 2016; Vermote et al., 2014) is used to generate the DNB surface bidirectional reflectance factor (BRF) using NASA's Level 1B calibrated radiance product as input (i.e., 6-minute granules, or 2366 km along track and ~3100 km across-track). Level 2G DNB surface reflectance is then generated by performing spatial and temporal aggregation to 15 arc-second grid cells over daily time periods (Campagnolo et al., 2016; Pahlevan et al., 2017; Wolfe et al., 1998; Yang and Wolfe, 2001). Daily Level 3 DNB BRDF/Albedo data are then retrieved using the heritage MODIS/VIIRS algorithm (MCD43/VNP43) (Liu et al., 2017, Liu et al., 2017; Wang et al., 2018), and corresponding snow flags are estimated using the VIIRS Normalized Difference Snow Index (NDSI) algorithm (VNP10) (Riggs et al., 2016, Riggs et al., 2017). The NDVI and NDSI values are used to determine the growing, dormant, and snow periods to routinely update the *a priori* global database of the DNB BRDF product (Cescatti et al., 2012; Liu et al., 2017, Liu et al., 2017; Román et al., 2009). Surface BRF from the VIIRS I1 (red) and I2 (NIR) channels is used to obtain daily estimates



of LAI (Knyazikhin et al., 1999; Park et al., 2017; Xiao et al., 2016). The retrieved LAI and clumping index values are then used to calculate the gap fraction probability ( $P_{gap}$ ). Note that the vegetation correction is not applied in Black Marble Collection 1 product as requested by the end users.

The nighttime branch describes the path followed to generate the final VNP46 products. We begin with the at-sensor TOA nighttime radiance (VNP46A1), along with the corresponding nighttime cloud mask, multiple solar/viewing/lunar geometry values (including moon-illuminated fraction and phase angles), and the daily snow and aerosol status flags. These science data sets (SDS) enable open access to the primary inputs used to generate NASA's Black Marble NTL time-series record, thus ensuring reproducibility of the final outputs. A series of temporal and spatial gap-filling techniques are also employed to improve the coverage of the VNP46 NTL product.

## **5 Scientific Data Sets (SDSs) from Black Marble Product Suite**

### **5.1 The VNP46A1/VJ146A1 Daily At-sensor TOA Nighttime Lights Product**

The daily at-sensor TOA nighttime lights product is available at 15 arc-second spatial resolution from January 2012 onward. VNP46A1/VJ146A1 product contains 26 SDS layers (Table 3) including sensor radiance, zenith and azimuth angles at-sensor, solar, and lunar, cloud mask flag, time, shortwave IR radiance, brightness temperatures, VIIRS quality flags, moon phase angle, and moon illumination fraction. Contents of VNP46A1/VJ146A1 product are given in List 1. Table 3 presents detailed information on the layers. Table 4 and Table 5 present the details of the flag description keys and quality flags (QF) of the VNP46A1/VJ146A1 product.

**List 1** Datasets in a sample of VNP46A1/VJ146A1 product.

- ~  HDFEOS
  - ~  ADDITIONAL
    -  FILE\_ATTRIBUTES
  - ~  GRIDS
    - ~  VNP\_Grid\_DNB
      - ~  Data Fields
        -  BrightnessTemperature\_M12
        -  BrightnessTemperature\_M13
        -  BrightnessTemperature\_M15
        -  BrightnessTemperature\_M16
        -  DNB\_At\_Sensor\_Radiance\_500m
        -  Glint\_Angle
        -  Granule
        -  Lunar\_Azimuth
        -  Lunar\_Zenith
        -  Moon\_Illumination\_Fraction
        -  Moon\_Phase\_Angle
        -  QF\_Cloud\_Mask
        -  QF\_DNB
        -  QF\_VIIRS\_M10
        -  QF\_VIIRS\_M11
        -  QF\_VIIRS\_M12
        -  QF\_VIIRS\_M13
        -  QF\_VIIRS\_M15
        -  QF\_VIIRS\_M16
        -  Radiance\_M10
        -  Radiance\_M11
        -  Sensor\_Azimuth
        -  Sensor\_Zenith
        -  Solar\_Azimuth
        -  Solar\_Zenith
        -  UTC\_Time
  - ~  HDFEOS INFORMATION
    -  StructMetadata.0

**Table 3** Scientific datasets included in the VNP46A1/VJ146A1 daily at-sensor TOA nighttime radiance product.

| Scientific Datasets<br>(SDS HDF Layers) | Units                                 | Description   | Bit Types | Fill<br>Value      | Valid<br>Range | Scale<br>Factor | Offset |
|-----------------------------------------|---------------------------------------|---------------|-----------|--------------------|----------------|-----------------|--------|
| DNB_At_Sensor_Radian                    | nW·cm <sup>-2</sup> ·sr <sup>-1</sup> | At-sensor DNB | 16-bit    | 65535 <sup>1</sup> | 0 - 65534      | 0.1             | 0.0    |

|                            |                                                 |                                    |                         |        |                |         |       |
|----------------------------|-------------------------------------------------|------------------------------------|-------------------------|--------|----------------|---------|-------|
| ce                         |                                                 | radiance                           | unsigned integer        |        |                |         |       |
| Sensor_Zenith              | Degrees                                         | Sensor zenith angle                | 16-bit signed integer   | -32768 | 0 – 9000       | 0.01    | 0.0   |
| Sensor_Azimuth             | Degrees                                         | Sensor azimuth angle               | 16-bit signed integer   | -32768 | -18000 – 18000 | 0.01    | 0.0   |
| Solar_Zenith               | Degrees                                         | Solar zenith angle                 | 16-bit signed integer   | -32768 | 0 – 18000      | 0.01    | 0.0   |
| Solar_azimuth              | Degrees                                         | Solar azimuth angle                | 16-bit signed integer   | -32768 | -18000 – 18000 | 0.01    | 0.0   |
| Lunar_Zenith               | Degrees                                         | Lunar zenith angle                 | 16-bit signed integer   | -32768 | 0 – 18000      | 0.01    | 0.0   |
| Lunar_Azimuth              | Degrees                                         | Lunar azimuth angle                | 16-bit signed integer   | -32768 | -18000 – 18000 | 0.01    | 0.0   |
| Glint_Angle                | Degrees                                         | Moon glint angle                   | 16-bit signed integer   | -32768 | -18000 – 18000 | 0.01    | 0.0   |
| UTC_Time                   | Decimal hours                                   | UTC time                           | 32-bit floating point   | -999.9 | 0 24           | 1.0     | 0.0   |
| QF_Cloud_Mask <sup>2</sup> | Unitless                                        | Cloud mask status                  | 16-bit unsigned integer | 65535  | 0 - 65534      | N/A     | N/A   |
| QF_DNB <sup>3</sup>        | Unitless                                        | DNB quality flag                   | 16-bit unsigned integer | 65535  | 0 - 65534      | N/A     | N/A   |
| Radiance_M10               | $W \cdot m^{-2} \cdot \mu m^{-1} \cdot sr^{-1}$ | Radiance in band M10               | 16-bit unsigned integer | 65535  | 0 - 65534      | 0.0013  | -0.04 |
| Radiance_M11               | $W \cdot m^{-2} \cdot \mu m^{-1} \cdot sr^{-1}$ | Radiance in band M11               | 16-bit unsigned integer | 65535  | 0 - 65534      | 0.00058 | -0.02 |
| BrightnessTemperature_M12  | Kelvins                                         | Brightness temperature of band M12 | 16-bit unsigned integer | 65535  | 0 - 65534      | 0.0025  | 203.0 |
| BrightnessTemperature_M13  | Kelvins                                         | Brightness temperature of band M13 | 16-bit unsigned integer | 65535  | 0 - 65534      | 0.0025  | 203.0 |
| BrightnessTemperature_M15  | Kelvins                                         | Brightness temperature of band M15 | 16-bit unsigned integer | 65535  | 0 - 65534      | 0.0041  | 111.0 |
| BrightnessTemperature_M16  | Kelvins                                         | Brightness temperature of          | 16-bit unsigned         | 65535  | 0 - 65534      | 0.0043  | 103.0 |

|                            |            |                            |                         |        |           |      |     |
|----------------------------|------------|----------------------------|-------------------------|--------|-----------|------|-----|
|                            |            | band M16                   | integer                 |        |           |      |     |
| QF_VIIRS_M10 <sup>4</sup>  | Unitless   | Quality flag of band M10   | 16-bit unsigned integer | 65535  | 0 - 65534 | N/A  | N/A |
| QF_VIIRS_M11 <sup>4</sup>  | Unitless   | Quality flag of band M11   | 16-bit unsigned integer | 65535  | 0 - 65534 | N/A  | N/A |
| QF_VIIRS_M12 <sup>4</sup>  | Unitless   | Quality flag of band M12   | 16-bit unsigned integer | 65535  | 0 - 65534 | N/A  | N/A |
| QF_VIIRS_M13 <sup>4</sup>  | Unitless   | Quality flag of band M13   | 16-bit unsigned integer | 65535  | 0 - 65534 | N/A  | N/A |
| QF_VIIRS_M15 <sup>4</sup>  | Unitless   | Quality flag of band M15   | 16-bit unsigned integer | 65535  | 0 - 65534 | N/A  | N/A |
| QF_VIIRS_M16 <sup>4</sup>  | Unitless   | Quality flag of band M16   | 16-bit unsigned integer | 65535  | 0 - 65534 | N/A  | N/A |
| Moon_Phase_Angle           | Degrees    | Moon phase angle           | 16-bit signed integer   | -32768 | 0 – 18000 | 0.01 | 0.0 |
| Moon_Illumination_Fraction | Percentage | Moon illumination fraction | 16-bit signed integer   | -32768 | 0 – 10000 | 0.01 | 0.0 |
| Granule                    | Unitless   | Number of selected Granule | 8-bit unsigned integer  | 255    | 0 - 254   | 1.0  | 0.0 |

<sup>1</sup> Note that the need to use fill-values can arise from various scenarios such as bad quality data or if the solar zenith angle < 108 degrees since that is the nighttime cut-off used in the code. <sup>2</sup> Details of QF\_Cloud\_Mask are shown in Table 4. <sup>3</sup> The scale and offset are for nighttime. Users should check the quality flags and metadata for specific values. <sup>4</sup> Details of QF\_DNB and QF for VIIRS band M10/11/12/13/15/16 are shown in Table 5.

**Table 4** Value of QF\_Cloud\_Mask in the VNP46A1/VJ146A1 product.

| Bit | Flag description key  | Interpretation       |
|-----|-----------------------|----------------------|
| 0   | Day/Night             | 0 = Night<br>1 = Day |
| 1-3 | Land/Water Background | 000 = Land & Desert  |

|     |                                                |                                                                                              |
|-----|------------------------------------------------|----------------------------------------------------------------------------------------------|
|     |                                                | 001 = Land no Desert<br>010 = Inland Water<br>011 = Sea Water<br>101 = Coastal               |
| 4-5 | Cloud Mask Quality                             | 00 = Poor<br>01 = Low<br>10 = Medium<br>11 = High                                            |
| 6-7 | Cloud Detection Results & Confidence Indicator | 00 = Confident Clear<br>01 = Probably Clear<br>10 = Probably Cloudy<br>11 = Confident Cloudy |
| 8   | Shadow Detected                                | 1 = Yes<br>0 = No                                                                            |
| 9   | Cirrus Detection (IR) (BTM15 – BTM16)          | 1 = Cloud<br>0 = No Cloud                                                                    |
| 10  | Snow/ Ice Surface                              | 1 = Snow/Ice<br>0 = No Snow/Ice                                                              |

**Table 5** Value of QF\_DNB and QF of VIIRS band M10/11/12/13/15/16 in the VNP46A1/VJ146A1 product.

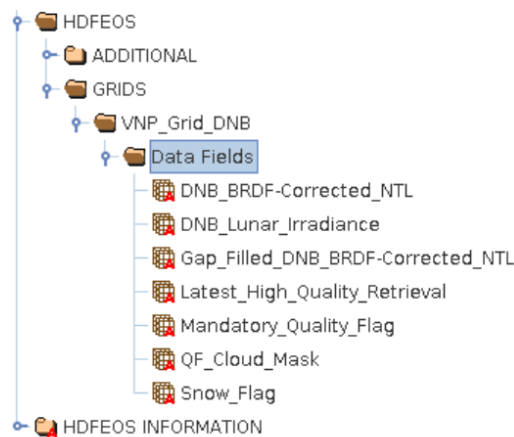
| SDS Layer                                                                                    | Flag Mask Values and Descriptions                                                                                                                                                                     |
|----------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| QF_DNB                                                                                       | 1 = Substitute_Cal<br>2 = Out_of_Range<br>4 = Saturation<br>8 = Temp_not_Nominal<br>16 = Stray_light<br>256 = Bowtie_Deleted/Range_bit<br>512 = Missing_EV<br>1024 = Cal_Fail<br>2048 = Dead_Detector |
| QF_VIIRS_M10<br>QF_VIIRS_M11<br>QF_VIIRS_M12<br>QF_VIIRS_M13<br>QF_VIIRS_M15<br>QF_VIIRS_M16 | 1 = Substitute_Cal<br>2 = Out_of_Range<br>4 = Saturation<br>8 = Temp_not_Nominal<br>256 = Bowtie_Deleted<br>512 = Missing_EV<br>1024 = Cal_Fail<br>2048 = Dead_Detector                               |

## 5.2 The VNP46A2/VJ146A2 Daily Moonlight-adjusted Nighttime Lights (NTL)

### Product

The daily moonlight and atmosphere-corrected NTL is available at 15 arc-second resolution from January 2012-present. The VNP46A2/VJ146A2 product has 7 layers containing information on BRDF-corrected NTL, Gap-filled BRDF-corrected NTL, lunar irradiance, mandatory quality flag, latest high-quality retrieval (number of days), snow flag, and cloud mask flag. Contents of VNP46A2/VJ146A2 product are given in List 2. The detailed VNP46A2/VJ146A2 layer properties are described in Table 6. Table 7 and Table 8 present the details of quality flags (QF) for the VNP46A2/VJ146A2 product.

**List 2** Datasets in a sample of VNP46A2/VJ146A2 product



**Table 6** Scientific datasets included in VNP46A2/VJ146A2 daily moonlight-adjusted NTL product.

| Scientific Data Sets<br>(SDS HDF Layers) | Units                                                     | Description                             | Bit Types                     | Fill Value | Valid Range   | Scale Factor | Offset |
|------------------------------------------|-----------------------------------------------------------|-----------------------------------------|-------------------------------|------------|---------------|--------------|--------|
| DNB_BRDF-Corrected_NTL                   | $\text{nWatts} \cdot \text{cm}^{-2} \cdot \text{sr}^{-1}$ | BRDF corrected<br>DNB NTL               | 16-bit<br>unsigned<br>integer | 65,535     | 0 –<br>65,534 | 0.1          | 0.0    |
| Gap_Filled_DNB_BRDF-Corrected_NTL        | $\text{nWatts} \cdot \text{cm}^{-2} \cdot \text{sr}^{-1}$ | Gap Filled BRDF<br>corrected DNB<br>NTL | 16-bit<br>unsigned<br>integer | 65,535     | 0 –<br>65,534 | 0.1          | 0.0    |
| DNB_Lunar_Irradiance                     | $\text{nWatts} \cdot \text{cm}^{-2}$                      | DNB Lunar<br>Irradiance                 | 16-bit<br>unsigned<br>integer | 65,535     | 0 –<br>65,534 | 0.1          | 0.0    |
| Mandatory_Quality_Flag <sup>1</sup>      | Unitless                                                  | Mandatory quality<br>flag               | 8-bit<br>unsigned<br>integer  | 255        | 0 – 3         | N/A          | N/A    |

|                               |                |                                                           |                         |        |            |     |     |
|-------------------------------|----------------|-----------------------------------------------------------|-------------------------|--------|------------|-----|-----|
| Latest_High_Quality_Retrieval | Number of days | Latest high quality BRDF corrected DNB radiance retrieval | 8-bit unsigned integer  | 255    | 0 – 254    | 1.0 | 0.0 |
| Snow_Flag <sup>2</sup>        | Unitless       | Flag for snow cover                                       | 8-bit unsigned integer  | 255    | 0 – 1      | N/A | N/A |
| QF_Cloud_Mask <sup>3</sup>    | Unitless       | Quality flag for cloud mask                               | 16-bit unsigned integer | 65,535 | 0 – 65,534 | N/A | N/A |

<sup>1</sup> Details of Mandatory\_Quality\_Flag are shown in Table 7. <sup>2</sup> Details of Snow\_Flag are shown in Table 8. <sup>3</sup> Details of QF\_Cloud\_Mask are shown in Table 4.

**Table 7** Values of the Mandatory\_Quality\_Flag in VNP46A2/VJ146A2 product.

| Value | Retrieval quality | Algorithm instance                                                      |
|-------|-------------------|-------------------------------------------------------------------------|
| 00    | High-quality      | Main algorithm (Persistent nighttime lights)                            |
| 01    | High-quality      | Main algorithm (Ephemeral Nighttime Lights)                             |
| 02    | Poor-quality      | Main algorithm (Outlier, potential cloud contamination or other issues) |
| 255   | No retrieval      | Fill value                                                              |

**Table 8** Values of the Snow\_Flag in VNP46A2/VJ146A2 product.

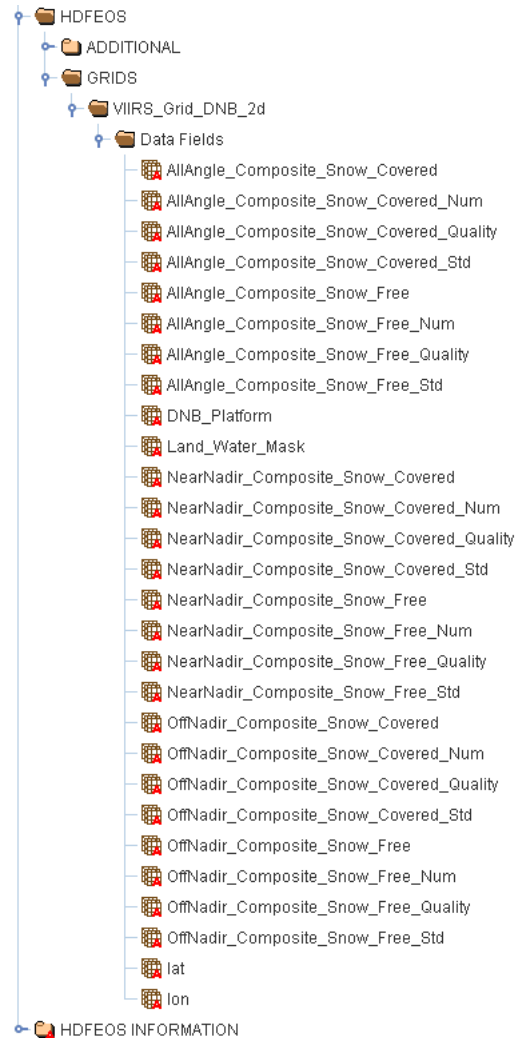
| Flag description key | Value | Interpretation |
|----------------------|-------|----------------|
| Snow/ Ice Surface    | 00    | No Snow/Ice    |
|                      | 01    | Snow/Ice       |
|                      | 255   | Fill value     |

### 5.3 The VNP46A3/VJ146A3 Monthly and VNP46A4/VJ146A4 Yearly Moonlight-adjusted Nighttime Lights (NTL) Product

The monthly and yearly moonlight and atmosphere-corrected NTL composite are available at 15 arc-second resolution from January 2012-present. The composite product has 28 layers containing information on NTL composite, the number of observations, quality, and standard deviation for multi-view zenith angle categories (near-nadir, off-nadir, and all angles) and snow status (snow-covered and snow-free) as well as land water mask, platform, latitude, and longitude. Contents of VNP46A3/A4 (VJ146A3/A4) product are given in List 2. The detailed VNP46A3/A4 (VJ146A3/A4) layer properties are

described in Table 6. Table 7 present the detailed description of quality flags (QF) for the VNP46A3/A4 (VJ146A3/A4) products.

### List 3 Datasets in a sample of VNP46A3/A4 (VJ146A3/A4) product



**Table 9** Scientific datasets included in VNP46A3/A4 (VJ146A3/A4) NTL composite products.

| Scientific Data Sets<br>(SDS HDF Layers) | Units                                     | Description                                                                   | Bit<br>Types            | Fill<br>Value | Valid<br>Range | Scale<br>Factor | Off<br>set |
|------------------------------------------|-------------------------------------------|-------------------------------------------------------------------------------|-------------------------|---------------|----------------|-----------------|------------|
| AllAngle_Composite_Snow_Covered          | nWatts·cm <sup>-2</sup> ·sr <sup>-1</sup> | Temporal Radiance Composite Using All Observations During Snow-covered Period | 16-bit unsigned integer | 65,535        | 0 – 65,534     | 0.1             | 0.0        |
| AllAngle_Composite_Snow_Covered_Num      | Number of observations                    | Number of Observations of                                                     | 16-bit unsigned         | 65,535        | 0 – 65,534     | 1.0             | 0.0        |



|                                                      |                                           |                                                                                                                            |                         |        |            |     |     |
|------------------------------------------------------|-------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|-------------------------|--------|------------|-----|-----|
|                                                      |                                           | Temporal Radiance Composite Using All Observations During Snow-covered Period                                              | integer                 |        |            |     |     |
| AllAngle_Composite_Snow_Covered_Quality <sup>1</sup> | Unitless                                  | Quality Flag of Temporal Radiance Composite Using All Observations During Snow-covered Period                              | 8-bit unsigned integer  | 255    | 0 – 254    | 1.0 | 0.0 |
| AllAngle_Composite_Snow_Covered_Std                  | nWatts·cm <sup>-2</sup> ·sr <sup>-1</sup> | Standard Deviation of Temporal Radiance Composite Using All Observations During Snow-covered Period                        | 16-bit unsigned integer | 65,535 | 0 – 65,534 | 0.1 | 0.0 |
| AllAngle_Composite_Snow_Free                         | nWatts·cm <sup>-2</sup> ·sr <sup>-1</sup> | Temporal Radiance Composite Using All Observations During Snow-free Period                                                 | 16-bit unsigned integer | 65,535 | 0 – 65,534 | 0.1 | 0.0 |
| AllAngle_Composite_Snow_Free_Num                     | Number of observations                    | Number of Observations of Temporal Radiance Composite Using All Observations During Snow-free Period                       | 16-bit unsigned integer | 65,535 | 0 – 65,534 | 1.0 | 0.0 |
| AllAngle_Composite_Snow_Free_Quality <sup>1</sup>    | Unitless                                  | Quality Flag of Temporal Radiance Composite Using All Observations During Snow-free Period                                 | 8-bit unsigned integer  | 255    | 0 – 254    | 1.0 | 0.0 |
| AllAngle_Composite_Snow_Free_Std                     | nWatts·cm <sup>-2</sup> ·sr <sup>-1</sup> | Standard Deviation of Temporal Radiance Composite Using All Observations During Snow-free Period                           | 16-bit unsigned integer | 65,535 | 0 – 65,534 | 0.1 | 0.0 |
| NearNadir_Composite_Snow_Covered                     | nWatts·cm <sup>-2</sup> ·sr <sup>-1</sup> | Temporal Radiance Composite Using Near Nadir Angle Observations (View Zenith Angle 0-20 degree) During Snow-covered Period | 16-bit unsigned integer | 65,535 | 0 – 65,534 | 0.1 | 0.0 |

|                                                       |                                           |                                                                                                                                                      |                         |        |            |     |     |
|-------------------------------------------------------|-------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|--------|------------|-----|-----|
| NearNadir_Composite_Snow_Covered_Num                  | Number of observations                    | Number of Observations of Temporal Radiance Composite Using Near Nadir Angle Observations (View Zenith Angle 0-20 degree) During Snow-covered Period | 16-bit unsigned integer | 65,535 | 0 – 65,534 | 1.0 | 0.0 |
| NearNadir_Composite_Snow_Covered_Quality <sup>1</sup> | Unitless                                  | Quality Flag of Temporal Radiance Composite Using Near Nadir Angle Observations (View Zenith Angle 0-20 degree) During Snow-covered Period           | 8-bit unsigned integer  | 255    | 0 – 254    | 1.0 | 0.0 |
| NearNadir_Composite_Snow_Covered_Std                  | nWatts·cm <sup>-2</sup> ·sr <sup>-1</sup> | Standard Deviation of Temporal Radiance Composite Using Near Nadir Angle Observations (View Zenith Angle 0-20 degree) During Snow-covered Period     | 16-bit unsigned integer | 65,535 | 0 – 65,534 | 0.1 | 0.0 |
| NearNadir_Composite_Snow_Free                         | nWatts·cm <sup>-2</sup> ·sr <sup>-1</sup> | Temporal Radiance Composite Using Near Nadir Angle Observations (View Zenith Angle 0-20 degree) During Snow-free Period                              | 16-bit unsigned integer | 65,535 | 0 – 65,534 | 0.1 | 0.0 |
| NearNadir_Composite_Snow_Free_Num                     | Number of observations                    | Number of Observations of Temporal Radiance Composite Using Near Nadir Angle Observations (View Zenith Angle 0-20 degree) During Snow-free Period    | 16-bit unsigned integer | 65,535 | 0 – 65,534 | 1.0 | 0.0 |

|                                                              |                                           |                                                                                                                                                                               |                               |        |               |     |     |
|--------------------------------------------------------------|-------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|--------|---------------|-----|-----|
| NearNadir<br>_Composite_Snow_Free_Qu<br>ality <sup>1</sup>   | Unitless                                  | Quality Flag of<br>Temporal Radiance<br>Composite Using Near<br>Nadir Angle<br>Observations (View<br>Zenith Angle 0-20<br>degree) During Snow-<br>free Period                 | 8-bit<br>unsigned<br>integer  | 255    | 0 –<br>254    | 1.0 | 0.0 |
| NearNadir<br>_Composite_Snow_Free_Std                        | nWatts·cm <sup>-2</sup> ·sr <sup>-1</sup> | Standard Deviation of<br>Temporal Radiance<br>Composite Using Near<br>Nadir Angle<br>Observations (View<br>Zenith Angle 0-20<br>degree) During Snow-<br>free Period           | 16-bit<br>unsigned<br>integer | 65,535 | 0 –<br>65,534 | 0.1 | 0.0 |
| OffNadir_Composite_Snow_<br>Covered                          | nWatts·cm <sup>-2</sup> ·sr <sup>-1</sup> | Temporal Radiance<br>Composite Using Off<br>Nadir Angle<br>Observations (View<br>Zenith Angle 40-60<br>degree) During Snow-<br>covered Period                                 | 16-bit<br>unsigned<br>integer | 65,535 | 0 –<br>65,534 | 0.1 | 0.0 |
| OffNadir<br>_Composite_Snow_Covered<br>_Num                  | Number of<br>observations                 | Number of<br>Observations of<br>Temporal Radiance<br>Composite Using Off<br>Nadir Angle<br>Observations (View<br>Zenith Angle 40-60<br>degree) During Snow-<br>covered Period | 16-bit<br>unsigned<br>integer | 65,535 | 0 –<br>65,534 | 1.0 | 0.0 |
| OffNadir<br>_Composite_Snow_Covered<br>_Quality <sup>1</sup> | Unitless                                  | Quality Flag of<br>Temporal Radiance<br>Composite Using Off<br>Nadir Angle<br>Observations (View<br>Zenith Angle 40-60<br>degree) During Snow-<br>covered Period              | 8-bit<br>unsigned<br>integer  | 255    | 0 –<br>254    | 1.0 | 0.0 |
| OffNadir                                                     | nWatts·cm <sup>-2</sup> ·sr <sup>-1</sup> | Standard Deviation of                                                                                                                                                         | 16-bit                        | 65,535 | 0 –           | 0.1 | 0.0 |

|                                                   |                                           |                                                                                                                                                   |                         |        |            |     |     |
|---------------------------------------------------|-------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|--------|------------|-----|-----|
| _Composite_Snow_Covered_Std                       |                                           | Temporal Radiance Composite Using Off Nadir Angle Observations (View Zenith Angle 40-60 degree) During Snow-covered Period                        | unsigned integer        |        | 65,534     |     |     |
| OffNadir_Composite_Snow_Free                      | nWatts·cm <sup>-2</sup> ·sr <sup>-1</sup> | Temporal Radiance Composite Using Off Nadir Angle Observations (View Zenith Angle 40-60 degree) During Snow-free Period                           | 16-bit unsigned integer | 65,535 | 0 – 65,534 | 0.1 | 0.0 |
| OffNadir_Composite_Snow_Free_Num                  | Number of observations                    | Number of Observations of Temporal Radiance Composite Using Off Nadir Angle Observations (View Zenith Angle 40-60 degree) During Snow-free Period | 16-bit unsigned integer | 65,535 | 0 – 65,534 | 1.0 | 0.0 |
| OffNadir_Composite_Snow_Free_Quality <sup>1</sup> | Unitless                                  | Quality Flag of Temporal Radiance Composite Using Off Nadir Angle Observations (View Zenith Angle 40-60 degree) During Snow-free Period           | 8-bit unsigned integer  | 255    | 0 – 254    | 1.0 | 0.0 |
| OffNadir_Composite_Snow_Free_Std                  | nWatts·cm <sup>-2</sup> ·sr <sup>-1</sup> | Standard Deviation of Temporal Radiance Composite Using Off Nadir Angle Observations (View Zenith Angle 40-60 degree) During Snow-free Period     | 16-bit unsigned integer | 65,535 | 0 – 65,534 | 0.1 | 0.0 |
| DNB_Platform <sup>2</sup>                         | Unitless                                  | Platform                                                                                                                                          | 8-bit unsigned integer  | 255    | 0 – 254    | 1.0 | 0.0 |

|                              |               |                 |                              |     |            |     |         |
|------------------------------|---------------|-----------------|------------------------------|-----|------------|-----|---------|
|                              |               |                 | integer                      |     |            |     |         |
| Land_Water_Mask <sup>3</sup> | Unitless      | Land Water Mask | 8-bit<br>unsigned<br>integer | 255 | 0 –<br>254 | 1.0 | 0.0     |
| Lat                          | Degrees_north | Latitude        | 64-bit<br>floating-<br>point | N/A | N/A        | N/A | N/<br>A |
| Lon                          | Degree_east   | Longitude       | 64-bit<br>floating-<br>point | N/A | N/A        | N/A | N/<br>A |

<sup>1</sup> Details of Quality\_Flag are shown in Table 10. <sup>2</sup> Details of DNB\_Platform are shown in Table 11. <sup>3</sup> Details of Land\_Water\_Mask are shown in Table 4.

**Table 10** Values of the Quality Flag in VNP46A3/A4 (VJ146A3/A4) product.

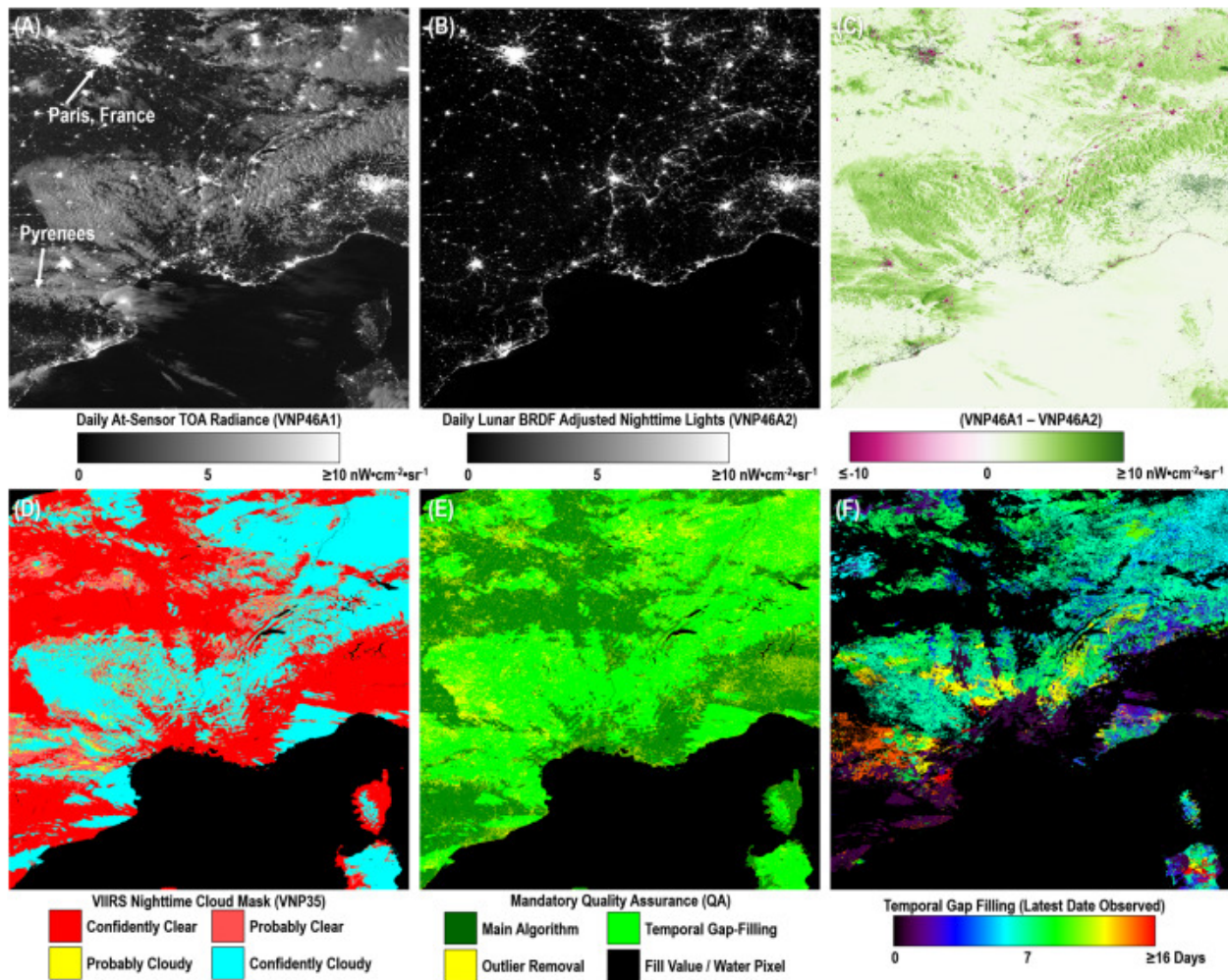
| Value | Retrieval quality | Algorithm instance                                                           |
|-------|-------------------|------------------------------------------------------------------------------|
| 00    | Good-quality      | The number of observations used for the composite is larger than 3           |
| 01    | Poor-quality      | The number of observations used for the composite is less than or equal to 3 |
| 02    | Gap filled        | Gap filled NTL based on historical data                                      |
| 255   | Fill value        | Fill value                                                                   |

**Table 11** Values of the DNB\_Platform in VNP46A3/A4 (VJ146A3/A4) product.

| Value | Sensors                        |
|-------|--------------------------------|
| 00    | Suomi-NPP                      |
| 01    | NOAA-20                        |
| 02    | Suomi-NPP and NOAA-20 combined |
| 255   | Fill value                     |

## 5.4 Examples of the Black Marble Product Suite

The Black Marble product suite is available both retrospectively, via NASA's Level 1 and Atmosphere Archive and Distribution System Distributed Active Archive Center (LAADS-DAAC), and in forward near-real-time (NRT) data streams, via NASA's Land, Atmosphere Near-Real-time Capability for EOS (LANCER) with a latency of about three hours. The NRT data are mainly used in response to disasters and other management applications, which require low latency data access.

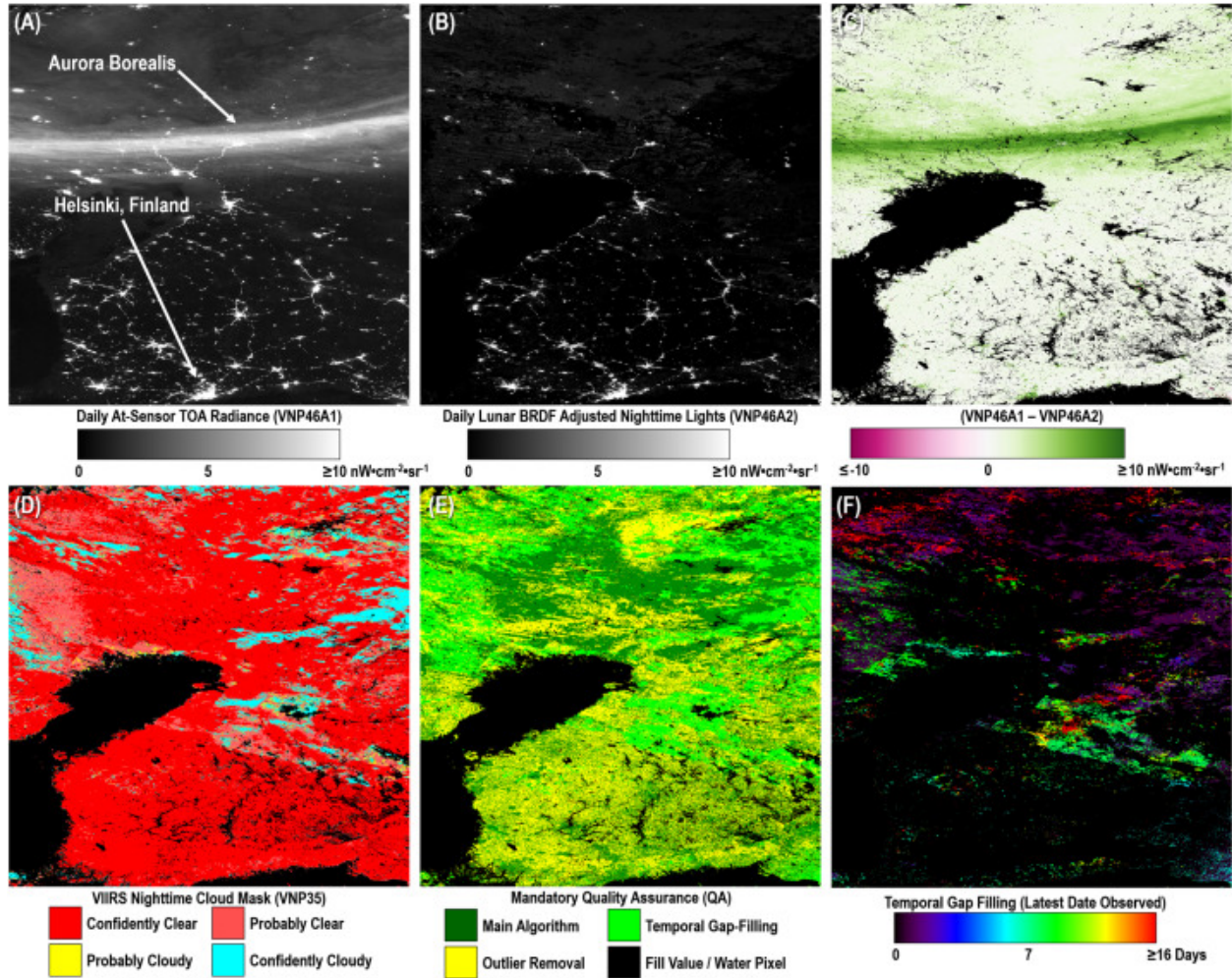


**Figure 4** Black Marble product suite components for a  $10^{\circ} \times 10^{\circ}$  Level 3 tile over France and the Balearic Sea region (h18v04; DOY 2015-091). The full-moon-illuminated and 51% cloud-contaminated scene illustrates the challenges of nighttime cloud masking over snow-covered surfaces (e.g., the French Alps and the Pyrenees).

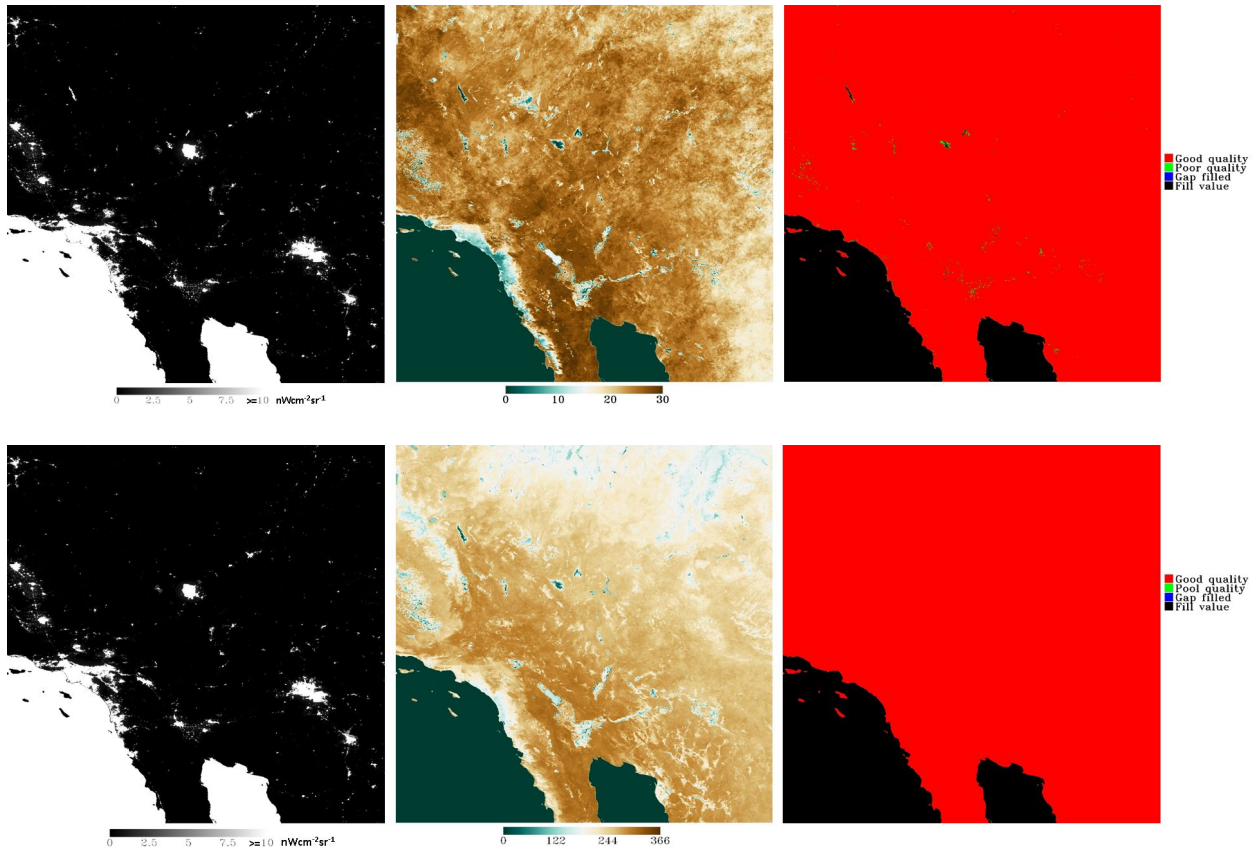
Figure 4 and Figure 5 illustrate the key processing steps used to retrieve high-quality NTL as part of NASA's Black Marble product suite. Cloud-free, atmospheric-, seasonal-, and moonlight BRDF-corrected DNB nighttime radiance is produced using the nighttime DNB-at-sensor radiance (VNP46A1), nighttime cloud mask, aerosol optical depth values, snow status flag, Ross-Li DNB BRDF model parameters and albedo values, Pgap, and per-pixel estimates of DNB lunar irradiance and corresponding geometries. A mandatory quality assurance (QA) flag is then provided to establish the pixel-specific estimates of retrieval performance. Note that when the temporal gap-filling routine is called upon, as reported in the mandatory quality assurance (QA) Flags (Table 7), the latest high-quality date observed (based on retrievals using the main algorithm) is reported as a separate SDS layer. If an outlier is still detected after temporal gap-filling, then the VNP46 algorithm defaults to a monthly climatology, based on the most recently available moonless high QA values. Thus, through judicious use of the VNP46 product quality flag, the end-user can establish whether a particular temporally gap-filled NTL value is based on a recent date or not. This results in a traceable, moonlight-adjusted, NTL product to assess current versus recent NTL conditions, while reducing persistent data gaps caused by nighttime clouds, snow, and other



ephemeral artifacts (e.g., the Aurora Borealis - cf., Figure 5). Figure 6 illustrates the monthly (VNP46A3) and yearly (VNP46A4) NTL composite, the associated the number of valid observations used for the composite, and quality flags generated from daily lunar-BRDF-corrected NTL product (VNP46A2) for tile h06v05 in 2016.



**Figure 5** VNP46 product suite components for a  $10^\circ \times 10^\circ$  Level 3 tile over Sweden and Finland (h20v02; DOY 2013-080). The half-moon-illuminated and 30% cloud-contaminated scene is shown to capture extraneous light emissions north of the Gulf of Bothnia caused by the Aurora Borealis.



**Figure 6** VNP46A3 monthly (upper) and VNP46A4 yearly (bottom) NTL composite (left), the associated number of observations for the composite (middle) and quality (right) for a  $10^{\circ} \times 10^{\circ}$  Level 3 tile h06v05 in 2016.

## 6 Evaluation and Validation of the Product

The overarching goal of NASA's Black Marble science product development effort is to achieve a “breakthrough” performance specification (cf., Table 9) by conducting the following: (1) long-term stability monitoring of the entire VNP46 algorithm processing chain, including the fundamental (Level 1B) VIIRS DNB time series record, terrain-corrected geolocation, stray light correction, and calibration LUTs; and (2) global quality assessment, uncertainty quantification, and product validation. To assess progress, we have developed a series of benchmark tests to quantify product performance at representative spatial and temporal scales. This comprehensive suite of benchmark tests and assessment metrics are meant to ensure that variations in VNP46 product performance can be identified quickly so that improvements can be implemented in a timely fashion. It also enables the end-user to consider the products in their appropriate context, e.g., by anticipating appropriate noise reduction levels under specific retrieval conditions.

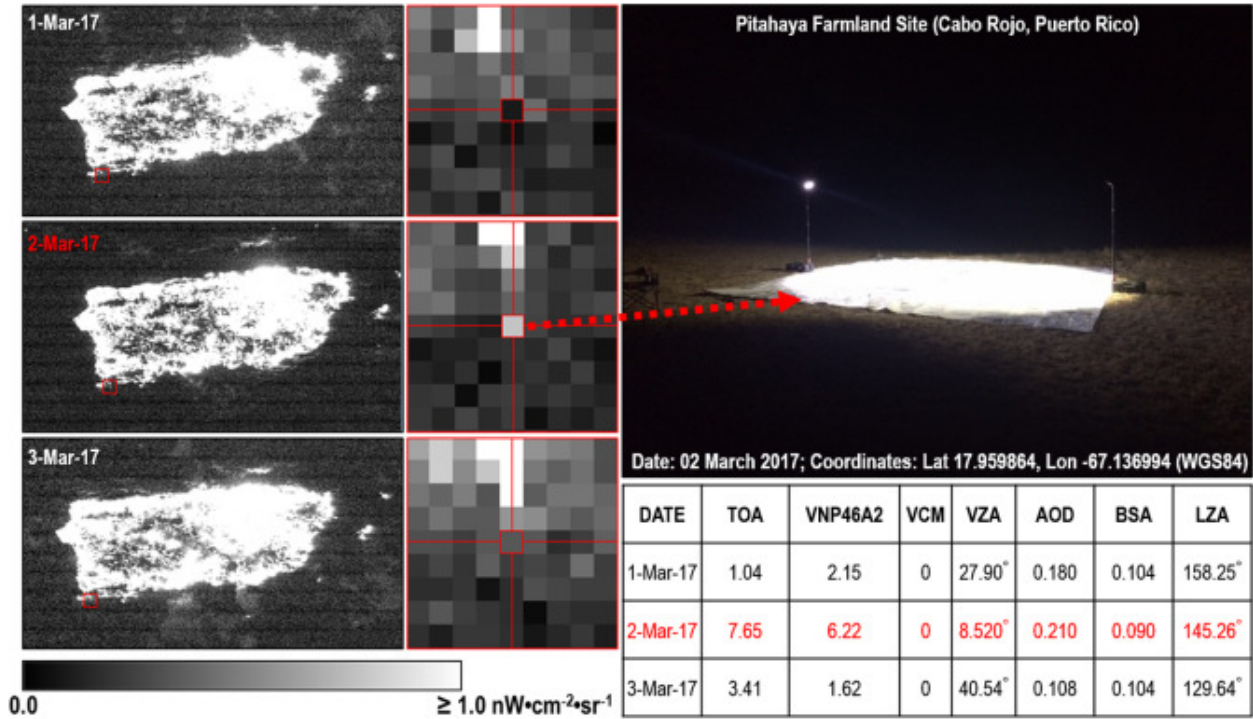


**Table 9** Key performance metrics established for NASA's Black Marble product suite.

| Key performance metrics    | Threshold                                                      | Breakthrough                                                    | Goal                                                            |
|----------------------------|----------------------------------------------------------------|-----------------------------------------------------------------|-----------------------------------------------------------------|
| NTL detection limit (Lmin) | $3.0 \text{ nW} \cdot \text{cm}^{-2} \cdot \text{sr}^{-1}$     | $0.5 \text{ nW} \cdot \text{cm}^{-2} \cdot \text{sr}^{-1}$      | $0.25 \text{ nW} \cdot \text{cm}^{-2} \cdot \text{sr}^{-1}$     |
| NTL robustness (L0)        | $\pm 3.0 \text{ nW} \cdot \text{cm}^{-2} \cdot \text{sr}^{-1}$ | $\pm 0.10 \text{ nW} \cdot \text{cm}^{-2} \cdot \text{sr}^{-1}$ | $\pm 0.05 \text{ nW} \cdot \text{cm}^{-2} \cdot \text{sr}^{-1}$ |
| Stray light error          | $0.45 \text{ nW} \cdot \text{cm}^{-2} \cdot \text{sr}^{-1}$    | $0.25 \text{ nW} \cdot \text{cm}^{-2} \cdot \text{sr}^{-1}$     | $< 0.1 \text{ nW} \cdot \text{cm}^{-2} \cdot \text{sr}^{-1}$    |
| Spatial resolution         | 742m ( $\pm 5\%$ )                                             | 500m ( $\pm 5\%$ )                                              | $\leq 200\text{m}$ ( $\pm 5\%$ )                                |
| Temporal resolution        | Monthly                                                        | Daily                                                           | Hourly                                                          |
| Geolocation uncertainty    | 133m                                                           | 50m                                                             | 20m                                                             |

A series of benchmark tests were designed to quantify errors inherited from the upstream products (i.e., VIIRS calibrated radiance, cloud mask, aerosol retrieval, etc.), provided a relative assessment of NTL product performance. The initial validation results are presented together with example case studies that are available in related publications in section 10. To establish the absolute accuracy of the final NTL retrievals, one must also assess the NTL products against an independent source of reference data. Unfortunately, quality-assessed *in situ* NTL measurements are not widely available; let alone, at the spatial and temporal densities necessary to capture the full range of retrieval conditions. Recent NASA Black Marble product validation efforts have therefore focused on developing guidelines for accuracy assessment of NTL products through a number of international initiatives.

Figure 7 shows an example of the accuracy assessment of NTL products through a field experiment at the Pitahaya farmland site in Cabo Rojo, PR. During the night of 2 March 2017, at 02:00 local time, the Puerto Rico's Working Group on Light Pollution (PRWGLP) team conducted a validation experiment at the Pitahaya site. A stable point source was reflected by a 30 m<sup>2</sup> Lambertian target to generate an in-band DNB radiance at sensor of  $\sim 0.45 \text{ nW} \cdot \text{cm}^{-2} \cdot \text{sr}^{-1}$ . Additional Sky-Quality Meter instrument data recordings (Falchi et al., 2016; Kyba et al., 2011, Kyba et al., 2013; Schnitt et al., 2013) with specialized filters matching the VIIRS relative spectral response, as well as atmospheric measurements from nearby Aerosol Robotic Network (AERONET) sun photometers (Holben et al., 1998) were used to characterize atmospheric conditions.



**Figure 7** The NTL radiance at the Pitahaya farmland site in Cabo Rojo, PR on 1st, 2nd and 3rd March 2017. The top-right image shows the setup of the stable point source. TOA and VNP46A2 values are in  $\text{nW}\cdot\text{cm}^{-2}\cdot\text{sr}^{-1}$ .  $\text{VCM} = 0$  represents cloud free overpasses. LZA is lunar zenith angle, and the values larger than  $108^\circ$  correspond to moonless nights.

The validation approach follows the assessment method first described in *Cao and Bai (2014)*, which relies on quantitative analysis and stability monitoring of stable light point sources. We used the following parameters to generate our radiative transfer calculations: (1) atmospheric transmittance = 0.8 (based on 6S radiative transfer code and AERONET calculations), a target reflectance = 0.8, and 16 W of total effective irradiance incident on the reflective surface. Results in Figure 7 also illustrate how the detected VIIRS at-sensor cloud-corrected radiance (or TOA) and VNP46A2 estimates over the pixel centered on the reflective point source were within the VNP46A2 product's “breakthrough” requirement specifications for the NTL detection limit,  $L_{\text{min}}$  ( $0.43 \text{ nW}\cdot\text{cm}^{-2}\cdot\text{sr}^{-1}$ ) after removing background noise measured the days prior and after activation of the stable light point sources. We found that the final VNP46A2 product resulted in a 16.95% sensitivity enhancement (due to reduced background noise), as confirmed in previous benchmark tests, compared to the at-sensor cloud-corrected radiance product (TOA) under observed moon-free conditions.

## 7 Data Archives

The VNP46 suite of operational products are archived and supported by NASA's LAADS DAAC data center <https://ladsweb.modaps.eosdis.nasa.gov/>.

The VNP46 suite of near-real-time products with a latency of about three hours are available at the NASA LANCE: NASA Near-Real-time Data and Imagery <https://earthdata.nasa.gov/earth-observation-data/near-real-time>.

## 8 Data Usage and Citation Policies

Please find detailed information about how to use and how to cite the data on the webpage [https://modaps.modaps.eosdis.nasa.gov/services/faq/LAADS\\_Data-Use\\_Citation\\_Policies.pdf](https://modaps.modaps.eosdis.nasa.gov/services/faq/LAADS_Data-Use_Citation_Policies.pdf).

## 9 Contact Information

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## 10 Related Publications

Román, M.O., Wang, Z., Sun, Q., Kalb, V., Miller, S.D., Molthan, A., Schultz, L., Bell, J., Stokes, E.C., Pandey, B. and Seto, K.C., et al. (2018). NASA's Black Marble nighttime lights product suite. *Remote Sens. Environ.* 210, 113-143. doi:10.1016/j.rse.2018.03.017.

Román, M.O. and Stokes, E.C. (2015). Holidays in lights: Tracking cultural patterns in demand for energy services. *Earth's Future*, 3, 182–205.

Román, M.O., Stokes, E.C., Shrestha, R., Wang, Z., Schultz, L., Sepúlveda Carlo, E.A., Sun, Q., Bell, J., Molthan, A., Kalb, V., Ji, C., Seto, K.C., McClain, S.N., Enenkel, M., 2019. Satellite-based assessment of electricity restoration efforts in Puerto Rico after Hurricane Maria. *PLoS One*. doi:10.1371/journal.pone.0218883

Wang, Z., Román, M.O., Kalb, V.L., Miller, S.D., Zhang, J., Shrestha, R.M., 2021. Quantifying uncertainties in nighttime light retrievals from Suomi-NPP and NOAA-20 VIIRS Day/Night Band data. *Remote Sensing of Environment* 263, 112557. <https://doi.org/10.1016/J.RSE.2021.112557>

Wang, Z., Shrestha, R.M., Roman, M.O., Kalb, V.L., 2022. NASA's Black Marble Multiangle Nighttime

Lights Temporal Composites. IEEE Geoscience and Remote Sensing Letters 19.  
<https://doi.org/10.1109/LGRS.2022.3176616>

Wang, Z., Román, M. O., Sun, Q., Molthan, A. L., Schultz, L. A., and Kalb, V. L. (2018). Monitoring Disaster-related Power Outages Using NASA Black Marble Nighttime Light Product. Int. Arch. Photogramm. Remote Sens. Spatial Inf. Sci., XLII-3, 1853-1856, <https://doi.org/10.5194/isprs-archives-XLII-3-1853-2018>, 2018.

Wang, Z., Shrestha, R. and Román, M. O., (2020). NASA's Black Marble Nighttime Lights Product Suite Algorithm Theoretical Basis Document (ATBD), Version 1.1, July 2020. Available in [https://viirsland.gsfc.nasa.gov/PDF/VIIRS\\_BlackMarble\\_ATBD\\_V1.1.pdf](https://viirsland.gsfc.nasa.gov/PDF/VIIRS_BlackMarble_ATBD_V1.1.pdf)

Cole, T.A., Wanik, D.W., Molthan, A.L., Román, M.O. and Griffin, R.E. (2017). Synergistic use of nighttime satellite data, electric utility infrastructure, and ambient population to improve power outage detections in urban areas. Remote Sensing, 9(3), 286. doi:10.3390/rs9030286.

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## Appendix A: Metadata (Attributes) in VNP46A1 Product

```
netcdf VNP46A1.A2013200.h10v04.001.2019115102717.h5 {

// global attributes:
    :Platform_Short_Name = "NPP" ;
    :ProductionTime = "2019-04-25 10:27:17.000" ;
    :ShortName = "VNP46A1" ;
    :PGEVersion = "1.0.8" ;
    :PGE_EndTime = "2013-07-19 23:59:59.000000Z" ;
    :HorizontalTileNumber = "10" ;
    :identifier_product_doi_authority = "http://dx.doi.org" ;
    :PGE_Name = "PGE554" ;
    :ProcessVersion = "001" ;
    :EndTime = "2013-07-19 23:59:59" ;
    :VerticalTileNumber = "04" ;
    :InputPointer =
"/MODAPSops4/archive/f7066/running/VNP_LP_L5lm7/1694463630/NPP_VMAES_L1.A2013200.0536.001.2016356233722.hdf/MODAPSops4/archiv
e/f7066/running/VNP_LP_L5lm7/1694463630/NPP_VMAES_L1.A2013200.0718.001.2016357002057.hdf/MODAPSops4/archive/f7066/running/VNP
_LP_L5lm7/1694463630/NPP_VMAES_L1.A2013200.0854.001.2016357005444.hdf/MODAPSops4/archive/f7066/running/VNP_LP_L5lm7/1694463630
/VNP35_L2.A2013200.0536.001.2016357002544.hdf/MODAPSops4/archive/f7066/running/VNP_LP_L5lm7/1694463630/VNP35_L2.A2013200.0718.0
01.2016357005809.hdf/MODAPSops4/archive/f7066/running/VNP_LP_L5lm7/1694463630/VNP35_L2.A2013200.0854.001.2016357013159.hdf/MOD
APSops4/archive/f7066/running/VNP_LP_L5lm7/1694463630/NPP_VDNES_L1.A2013200.0536.001.2016356233722.hdf/MODAPSops4/archive/f7066/
running/VNP_LP_L5lm7/1694463630/NPP_VDNES_L1.A2013200.0718.001.2016357002057.hdf/MODAPSops4/archive/f7066/running/VNP_LP_L5lm
7/1694463630/NPP_VDNES_L1.A2013200.0854.001.2016357005444.hdf" ;
    :LongName = "VIIRS/NPP Daily Gridded Day Night Band Linear Lat Lon Grid Night" ;
    :AlgorithmType = "OPS" ;
    :StartTime = "2013-07-19 00:00:00" ;
    :InstrumentShortname = "VIIRS" ;
    :identifier_product_doi = "10.5067/VIIRS/VNP46A1.001" ;
    :SatelliteInstrument = "NPP_OPS" ;
    :LocalGranuleID = "VNP46A1.A2013200.h10v04.001.2019115102717.h5" ;
    :TileID = "61010004" ;
    :ProcessingEnvironment = "Linux minion7066 3.10.0-957.5.1.el7.x86_64 #1 SMP Fri Feb 1 14:54:57 UTC 2019 x86_64 x86_64 x86_64
GNU/Linux" ;
    :NumberofInputGranules = "3" ;
    :PGE_StartTime = "2013-07-19 00:00:00.000" ;

group: HDFEOS {

    group: ADDITIONAL {

        group: FILE_ATTRIBUTES {
        } // group FILE_ATTRIBUTES
    } // group ADDITIONAL

    group: GRIDS {

        group: VNP_Grid_DNB {

            // group attributes:
            :InputPointer_CM =
"NPP_CMN.data.h10v04.A2013200.0536.hdf:NPP_CMN.data.h10v04.A2013200.0718.hdf:NPP_CMN.data.h10v04.A2013200.0854.hdf" ;
            :InputPointer_L2G_DNB =
"NPP_DNBn.data.h10v04.A2013200.0536.hdf:NPP_DNBn.data.h10v04.A2013200.0718.hdf:NPP_DNBn.data.h10v04.A2013200.0854.hdf" ;
            :InputPointer_L2G_Ang =
"NPP_DNBn.angles.h10v04.A2013200.0536.hdf:NPP_DNBn.angles.h10v04.A2013200.0718.hdf:NPP_DNBn.angles.h10v04.A2013200.0854.hdf" ;
            :InputPointer_L2G_PNTR =
"NPP_PTDN.h10v04.A2013200.0536.hdf:NPP_PTDN.h10v04.A2013200.0718.hdf:NPP_PTDN.h10v04.A2013200.0854.hdf" ;
            :InputPointer_DNB =
"/MODAPSops4/archive/f7066/running/VNP_LP_L5lm7/1694463630/NPP_VDNES_L1.A2013200.0536.001.2016356233722.hdf/MODAPSops4/archiv
e/f7066/running/VNP_LP_L5lm7/1694463630/NPP_VDNES_L1.A2013200.0718.001.2016357002057.hdf/MODAPSops4/archive/f7066/running/VNP
_LP_L5lm7/1694463630/NPP_VDNES_L1.A2013200.0854.001.2016357005444.hdf" ;
            :InputPointer_Mod =
"NPP_MOD.data.h10v04.A2013200.0536.hdf:NPP_MOD.data.h10v04.A2013200.0718.hdf:NPP_MOD.data.h10v04.A2013200.0854.hdf" ;
            :useCM = "No" ;
            :CMfill = "No" ;
            :RangeBeginningDate = "2013-07-19" ;
            :RangeBeginningTime = "00:00:00" ;
            :RangeEndingDate = "2013-07-19" ;
            :RangeEndingTime = "23:59:59" ;
            :NorthBoundingCoord = 50. ;
            :SouthBoundingCoord = 40. ;
```

```

:EastBoundingCoord = -70. ;
:WestBoundingCoord = -80. ;
:TileID = 61010004 ;
:TileMode = "Night" ;
:HorizontalTileNumber = 10 ;
:VerticalTileNumber = 4 ;
:NumberofInputGranules = 3 ;
:PGE_Name = "PGE554" ;
:PGEVersion = "1.0.8" ;
:SatelliteInstrument = "NPP_OPS" ;
:ReprocessingPlanned = "metadata field" ;
:ReprocessingActual = "metadata field" ;
:ProcessingEnvironment = "Linux minion7066 3.10.0-957.5.1.el7.x86_64 #1 SMP Fri Feb 1 14:54:57 UTC 2019 x86_64 x86_64 x86_64
GNU/Linux" ;
:ScienceQualityFlagExplanation = "unknown" ;

```

```

group: Data\ Fields {
  dimensions:
    phony_dim_0 = 2400 ;
  variables:
    ushort BrightnessTemperature_M12(phony_dim_0, phony_dim_0) ;
      BrightnessTemperature_M12:valid_min = 0 ;
      BrightnessTemperature_M12:valid_max = 65534 ;
      BrightnessTemperature_M12:_FillValue = 65535US ;
      BrightnessTemperature_M12:long_name = "Brightness Temperature of band M12 " ;
      BrightnessTemperature_M12:units = "Kelvins" ;
      BrightnessTemperature_M12:scale_factor = 0.0025f ;
      BrightnessTemperature_M12:add_offset = 203.f ;
    ushort BrightnessTemperature_M13(phony_dim_0, phony_dim_0) ;
      BrightnessTemperature_M13:valid_min = 0 ;
      BrightnessTemperature_M13:valid_max = 65534 ;
      BrightnessTemperature_M13:_FillValue = 65535US ;
      BrightnessTemperature_M13:long_name = "Brightness Temperature of band M13 " ;
      BrightnessTemperature_M13:units = "Kelvins" ;
      BrightnessTemperature_M13:scale_factor = 0.0025f ;
      BrightnessTemperature_M13:add_offset = 203.f ;
    ushort BrightnessTemperature_M15(phony_dim_0, phony_dim_0) ;
      BrightnessTemperature_M15:valid_min = 0 ;
      BrightnessTemperature_M15:valid_max = 65534 ;
      BrightnessTemperature_M15:_FillValue = 65535US ;
      BrightnessTemperature_M15:long_name = "Brightness Temperature of band M15 " ;
      BrightnessTemperature_M15:units = "Kelvins" ;
      BrightnessTemperature_M15:scale_factor = 0.0041f ;
      BrightnessTemperature_M15:add_offset = 111.f ;
    ushort BrightnessTemperature_M16(phony_dim_0, phony_dim_0) ;
      BrightnessTemperature_M16:valid_min = 0 ;
      BrightnessTemperature_M16:valid_max = 65534 ;
      BrightnessTemperature_M16:_FillValue = 65535US ;
      BrightnessTemperature_M16:long_name = "Brightness Temperature of band M16 " ;
      BrightnessTemperature_M16:units = "Kelvins" ;
      BrightnessTemperature_M16:scale_factor = 0.0043f ;
      BrightnessTemperature_M16:add_offset = 103.f ;
    ushort DNB_At_Sensor_Radiance_500m(phony_dim_0, phony_dim_0) ;
      DNB_At_Sensor_Radiance_500m:valid_min = 0 ;
      DNB_At_Sensor_Radiance_500m:valid_max = 65534 ;
      DNB_At_Sensor_Radiance_500m:long_name = "DNB at Sensor Radiance" ;
      DNB_At_Sensor_Radiance_500m:_FillValue = 65535US ;
      DNB_At_Sensor_Radiance_500m:units = "nW/(cm2 sr)" ;
      DNB_At_Sensor_Radiance_500m:scale_factor = 0.1f ;
      DNB_At_Sensor_Radiance_500m:add_offset = 0.f ;
    short Glint_Angle(phony_dim_0, phony_dim_0) ;
      Glint_Angle:valid_min = -18000 ;
      Glint_Angle:valid_max = 18000 ;
      Glint_Angle:_FillValue = -32768s ;
      Glint_Angle:long_name = "Glint Angle" ;
      Glint_Angle:units = "degrees" ;
      Glint_Angle:scale_factor = 0.01f ;
      Glint_Angle:add_offset = 0.f ;
    ubyte Granule(phony_dim_0, phony_dim_0) ;
      Granule:valid_min = 0 ;
      Granule:valid_max = 254 ;
      Granule:_FillValue = 255UB ;
      Granule:long_name = "Number of selected Granule" ;
      Granule:units = "none" ;
      Granule:scale_factor = 1.f ;

```

```

Granule:add_offset = 0.f;
short Lunar_Azimuth(phony_dim_0, phony_dim_0);
    Lunar_Azimuth:valid_min = -18000;
    Lunar_Azimuth:valid_max = 18000;
    Lunar_Azimuth: FillValue = -32768s;
    Lunar_Azimuth:long_name = "Lunar Azimuth Angle";
    Lunar_Azimuth:units = "degrees";
    Lunar_Azimuth:scale_factor = 0.01f;
    Lunar_Azimuth:add_offset = 0.f;
short Lunar_Zenith(phony_dim_0, phony_dim_0);
    Lunar_Zenith:valid_min = 0;
    Lunar_Zenith:valid_max = 18000;
    Lunar_Zenith: FillValue = -32768s;
    Lunar_Zenith:long_name = "Lunar Zenith Angle";
    Lunar_Zenith:units = "degrees";
    Lunar_Zenith:scale_factor = 0.01f;
    Lunar_Zenith:add_offset = 0.f;
short Moon_Illumination_Fraction(phony_dim_0, phony_dim_0);
    Moon_Illumination_Fraction:valid_min = 0;
    Moon_Illumination_Fraction:valid_max = 10000;
    Moon_Illumination_Fraction: FillValue = -32768s;
    Moon_Illumination_Fraction:long_name = "Moon Illumination Fraction";
    Moon_Illumination_Fraction:units = "percentage";
    Moon_Illumination_Fraction:scale_factor = 0.01f;
    Moon_Illumination_Fraction:add_offset = 0.f;
short Moon_Phase_Angle(phony_dim_0, phony_dim_0);
    Moon_Phase_Angle:valid_min = 0;
    Moon_Phase_Angle:valid_max = 18000;
    Moon_Phase_Angle: FillValue = -32768s;
    Moon_Phase_Angle:long_name = "Moon Phase Angle";
    Moon_Phase_Angle:units = "degrees";
    Moon_Phase_Angle:scale_factor = 0.01f;
    Moon_Phase_Angle:add_offset = 0.f;
ushort QF_Cloud_Mask(phony_dim_0, phony_dim_0);
    QF_Cloud_Mask:valid_min = 0;
    QF_Cloud_Mask:valid_max = 65534;
    QF_Cloud_Mask: FillValue = 65535US;
    QF_Cloud_Mask:long_name = "Cloud Mask Status";
    QF_Cloud_Mask:units = "class flags";
    QF_Cloud_Mask:flag_meanings = "bit 0: 0=Night, 1=Day\n bits 1-3: Land/Water 000=Land & Desert, 001=Land & no Desert, 010=Inland
Water, 011=Sea Water, 101=Coastal\n bits 4-5: Cloud Mask Quality 00=Poor, 01=Low, 10=Medium, 11=High\n bits 6-7: Cloud Confidence 00=Confident
Clear, 01=Probably Clear, 10=Probably Cloudy, 11=Confident Cloudy\n bit 8: Shadow Detected 1=Yes, 0=No\n bit 9: Cirrus Detection (IR) 1=Cloud,
0=No Cloud\n bit 10: Snow/Ice 1=Snow/Ice, 0=No Snow/Ice\n";
ushort QF_DNB(phony_dim_0, phony_dim_0);
    QF_DNB:valid_min = 0;
    QF_DNB:valid_max = 65534;
    QF_DNB: FillValue = 65535US;
    QF_DNB:long_name = "DNB QF";
    QF_DNB:units = "class flags";
ushort QF_VIIRS_M10(phony_dim_0, phony_dim_0);
    QF_VIIRS_M10:valid_min = 0;
    QF_VIIRS_M10:valid_max = 65534;
    QF_VIIRS_M10: FillValue = 65535US;
    QF_VIIRS_M10:long_name = "Quality Flag of Band M10";
    QF_VIIRS_M10:units = "class flags";
ushort QF_VIIRS_M11(phony_dim_0, phony_dim_0);
    QF_VIIRS_M11:valid_min = 0;
    QF_VIIRS_M11:valid_max = 65534;
    QF_VIIRS_M11: FillValue = 65535US;
    QF_VIIRS_M11:long_name = "Quality Flag of Band M11";
    QF_VIIRS_M11:units = "class flags";
ushort QF_VIIRS_M12(phony_dim_0, phony_dim_0);
    QF_VIIRS_M12:valid_min = 0;
    QF_VIIRS_M12:valid_max = 65534;
    QF_VIIRS_M12: FillValue = 65535US;
    QF_VIIRS_M12:long_name = "Quality Flag of Band M12";
    QF_VIIRS_M12:units = "class flags";
ushort QF_VIIRS_M13(phony_dim_0, phony_dim_0);
    QF_VIIRS_M13:valid_min = 0;
    QF_VIIRS_M13:valid_max = 65534;
    QF_VIIRS_M13: FillValue = 65535US;
    QF_VIIRS_M13:long_name = "Quality Flag of Band M13";
    QF_VIIRS_M13:units = "class flags";
ushort QF_VIIRS_M15(phony_dim_0, phony_dim_0);
    QF_VIIRS_M15:valid_min = 0;

```

```

        QF_VIIRS_M15:valid_max = 65534 ;
        QF_VIIRS_M15: FillValue = 65535US ;
        QF_VIIRS_M15:long_name = "Quality Flag of Band M15" ;
        QF_VIIRS_M15:units = "class flags" ;
    ushort QF_VIIRS_M16(phony_dim_0, phony_dim_0) ;
        QF_VIIRS_M16:valid_min = 0 ;
        QF_VIIRS_M16:valid_max = 65534 ;
        QF_VIIRS_M16: FillValue = 65535US ;
        QF_VIIRS_M16:long_name = "Quality Flag of Band M16" ;
        QF_VIIRS_M16:units = "class flags" ;
    ushort Radiance_M10(phony_dim_0, phony_dim_0) ;
        Radiance_M10:valid_min = 0 ;
        Radiance_M10:valid_max = 65534 ;
        Radiance_M10: FillValue = 65535US ;
        Radiance_M10:long_name = "Band M10 Radiance" ;
        Radiance_M10:units = "W/(m2 micron sr)" ;
        Radiance_M10:scale_factor = 0.0013f ;
        Radiance_M10:add_offset = -0.04f ;
    ushort Radiance_M11(phony_dim_0, phony_dim_0) ;
        Radiance_M11:valid_min = 0 ;
        Radiance_M11:valid_max = 65534 ;
        Radiance_M11: FillValue = 65535US ;
        Radiance_M11:long_name = "Band M11 Radiance" ;
        Radiance_M11:units = "W/(m2 micron sr)" ;
        Radiance_M11:scale_factor = 0.00058f ;
        Radiance_M11:add_offset = -0.02f ;
    short Sensor_Azimuth(phony_dim_0, phony_dim_0) ;
        Sensor_Azimuth:valid_min = -18000 ;
        Sensor_Azimuth:valid_max = 18000 ;
        Sensor_Azimuth: FillValue = -32768s ;
        Sensor_Azimuth:long_name = "Sensor Azimuth Angle" ;
        Sensor_Azimuth:units = "degrees" ;
        Sensor_Azimuth:scale_factor = 0.01f ;
        Sensor_Azimuth:add_offset = 0.f ;
    short Sensor_Zenith(phony_dim_0, phony_dim_0) ;
        Sensor_Zenith:valid_min = 0 ;
        Sensor_Zenith:valid_max = 9000 ;
        Sensor_Zenith: FillValue = -32768s ;
        Sensor_Zenith:long_name = "Sensor Zenith Angle" ;
        Sensor_Zenith:units = "degrees" ;
        Sensor_Zenith:scale_factor = 0.01f ;
        Sensor_Zenith:add_offset = 0.f ;
    short Solar_Azimuth(phony_dim_0, phony_dim_0) ;
        Solar_Azimuth:valid_min = -18000 ;
        Solar_Azimuth:valid_max = 18000 ;
        Solar_Azimuth: FillValue = -32768s ;
        Solar_Azimuth:long_name = "Solar Azimuth Angle" ;
        Solar_Azimuth:units = "degrees" ;
        Solar_Azimuth:scale_factor = 0.01f ;
        Solar_Azimuth:add_offset = 0.f ;
    short Solar_Zenith(phony_dim_0, phony_dim_0) ;
        Solar_Zenith:valid_min = 0 ;
        Solar_Zenith:valid_max = 18000 ;
        Solar_Zenith: FillValue = -32768s ;
        Solar_Zenith:long_name = "Solar Zenith Angle" ;
        Solar_Zenith:units = "degrees" ;
        Solar_Zenith:scale_factor = 0.01f ;
        Solar_Zenith:add_offset = 0.f ;
    float UTC_Time(phony_dim_0, phony_dim_0) ;
        UTC_Time:valid_min = 0 ;
        UTC_Time:valid_max = 24 ;
        UTC_Time: FillValue = -999.9f ;
        UTC_Time:long_name = "View Time (UTC)" ;
        UTC_Time:units = "decimal hours" ;
        UTC_Time:scale_factor = 1.f ;
        UTC_Time:add_offset = 0.f ;
} // group Data\Fields
} // group VNP_Grid_DNB
} // group GRIDS
} // group HDFEOS

group: HDFEOS\ INFORMATION {
    variables:
        string StructMetadata.0 ;

```



```
// group attributes:
:HDFEOSVersion = "HDFEOS_5.1.15" ;
} // group HDFEOS\ INFORMATION
}
```

## Appendix B: Metadata (Attributes) in VNP46A2 Product

```
netcdf VNP46A2.A2013200.h34v13.001.2020155060713.h5 {
```

```
// global attributes:
:SatelliteInstrument = "NPP OPS" ;
:DayNightFlag = "" ;
:PGENumber = "555" ;
:LongName = "VIIRS/NPP Gap-Filled Lunar BRDF-Adjusted Nighttime Lights Daily L3 Global Linear Lat Lon Grid" ;
:RangeBeginningTime = "00:00:00.000" ;
:NorthBoundingCoord = -40.f ;
:VersionID = "001" ;
:RangeEndingDate = "2013-07-19" ;
:PGE_StartTime = "2013-07-19 00:00:00.000" ;
:StartTime = "2013-07-19 00:00:00.000" ;
:LocalGranuleID = "VNP46A2.A2013200.h34v13.001.2020155060713.h5" ;
:ProductionTime = "2020-06-03 06:07:13.000" ;
:GRingPointLongitude = 160., 160., 170., 170. ;
:PlatformShortName = "NPP" ;
:identifier_product_doi_authority = "http://dx.doi.org" ;
:EndTime = "2013-07-19 23:59:59.000" ;
:VerticalTileNumber = "13" ;
:PGE_Name = "PGE555" ;
:InputPointer =
"VNPLG09GA.A2013200.h34v13.001.2020154195322.h5,VNPLG43DNBA1.A2013200.h34v13.001.2020155040500.h5,MCD12Q1.A2013001.Global.0
05.T1.Geo.h34v13.bin,MCD12Q1.A2013001.Global.005.T3.Geo.h34v13.bin,VNP46A1.A2013200.h34v13.001.2019115103737.h5,VNP04LGA.A201320
0.h34v13.001.2020155060525.hdf" ;
:RangeBeginningDate = "2013-07-19" ;
:SensorShortname = "VIIRS" ;
:ProcessingEnvironment = "Linux minion7414 3.10.0-1062.12.1.el7.x86_64 #1 SMP Tue Feb 4 23:02:59 UTC 2020 x86_64 x86_64 x86_64
GNU/Linux" ;
:TileID = "61034013" ;
:GRingPointLatitude = -50., -40., -40., -50. ;
:AlgorithmType = "SCI" ;
:PGE_EndTime = "2013-07-19 23:59:59.000" ;
:SouthBoundingCoord = -50.f ;
:LSIPS_AlgorithmVersion = "NPP_PR46A2 1.0.3" ;
:GranuleDayNightFlag = "" ;
:DataResolution = "Moderate" ;
:ProcessingCenter = "LandSIPS" ;
:HorizontalTileNumber = "34" ;
:PGEVersion = "1.0.6" ;
:ShortName = "VNP46A2" ;
:EastBoundingCoord = 170.f ;
:WestBoundingCoord = 160.f ;
:identifier_product_doi = "10.5067/VIIRS/VNP46A2.001" ;
:RangeEndingTime = "23:59:59.000" ;

group: HDFEOS {
  group: ADDITIONAL {
    group: FILE_ATTRIBUTES {
    } // group FILE_ATTRIBUTES
  } // group ADDITIONAL
group: GRIDS {
  group: VNP_Grid_DNB {
    group: Data\ Fields {
      dimensions:
        phony_dim_0 = 2400 ;
      variables:
        ushort DNB_BRDF-Corrected_NTL(phony_dim_0, phony_dim_0) ;
        DNB_BRDF-Corrected_NTL: FillValue = 65535US ;
        DNB_BRDF-Corrected_NTL:long_name = "BRDF Corrected DNB Radiance" ;
        DNB_BRDF-Corrected_NTL:units = "nWatts/(cm^2 sr) \n" ;
        DNB_BRDF-Corrected_NTL:valid_range = "0-65534 \n" ;
        DNB_BRDF-Corrected_NTL:offset = 0. ;
        DNB_BRDF-Corrected_NTL:scale_factor = 0.1 ;
```

```

ushort DNB_Lunar_Irradiance(phony_dim_0, phony_dim_0);
    DNB_Lunar_Irradiance: FillValue = 65535US;
    DNB_Lunar_Irradiance:long_name = "Lunar Irradiance";
    DNB_Lunar_Irradiance:units = "nWatts/cm^2 \n";
    DNB_Lunar_Irradiance:valid_range = "0-65534 \n";
    DNB_Lunar_Irradiance:offset = 0.;
    DNB_Lunar_Irradiance:scale_factor = 0.1;
ushort Gap_Filled_DNB_BRDF-Corrected_NTL(phony_dim_0, phony_dim_0);
    Gap_Filled_DNB_BRDF-Corrected_NTL: FillValue = 65535US;
    Gap_Filled_DNB_BRDF-Corrected_NTL:long_name = "Gap Filled BRDF Corrected DNB Radiance";
    Gap_Filled_DNB_BRDF-Corrected_NTL:units = "nWatts/(cm^2 sr) \n";
    Gap_Filled_DNB_BRDF-Corrected_NTL:valid_range = "0-65534 \n";
    Gap_Filled_DNB_BRDF-Corrected_NTL:offset = 0.;
    Gap_Filled_DNB_BRDF-Corrected_NTL:scale_factor = 0.1;
ubyte Latest_High_Quality_Retrieval(phony_dim_0, phony_dim_0);
    Latest_High_Quality_Retrieval: FillValue = 255UB;
    Latest_High_Quality_Retrieval:long_name = "The Latest High Quality BRDF Corrected DNB Radiance Retrieval";
    Latest_High_Quality_Retrieval:units = "Number of Days \n";
    Latest_High_Quality_Retrieval:valid_range = "0 - 254 \n";
    Latest_High_Quality_Retrieval:scale_factor = 1.;
    Latest_High_Quality_Retrieval:offset = 0.;
ubyte Mandatory_Quality_Flag(phony_dim_0, phony_dim_0);
    Mandatory_Quality_Flag: FillValue = 255UB;
    Mandatory_Quality_Flag:long_name = "Mandatory Quality Flag of BRDF Corrected DNB Radiance";
    Mandatory_Quality_Flag:units = "Unitless \n";
    Mandatory_Quality_Flag:valid_range = "0 - 3 \n";
    Mandatory_Quality_Flag:Description = "00\tHigh-Quality\tMain Algorithm (Persistent Nighttime Lights)\n01\tHigh-Quality\tMain
Algorithm (Ephemeral Nighttime Lights)\n02\tPoor-Quality\tMain Algorithm (Outlier, Potential cloud contamination or other issues)\n255\tNo
Retrieval\tFill Value \n";
    ushort QF_Cloud_Mask(phony_dim_0, phony_dim_0);
        QF_Cloud_Mask: FillValue = 65535US;
        QF_Cloud_Mask:long_name = "Cloud Mask Status";
        QF_Cloud_Mask:units = "Unitless \n";
        QF_Cloud_Mask:valid_range = "0 - 65534 \n";
        QF_Cloud_Mask:Description = "bit Flag description key: \n0\t Day/Night \n 0=Night 1=Day \n1-3\t Land/Water Background
\n\t000=Land & Desert \n\t001=Land no Desert \n\t010=Inland Water \n\t011=Sea Water \n\t101=Coastal \n4-5\t Cloud Mask Quality
\n\t00=Poor\n\t01=Low \n\t10=Medium \n\t11=High \n6-7\t Cloud Detection Results & Confidence Indicator \n\t00=Confident Clear
\n\t01=Probably Clear \n\t10=Probably Cloudy \n\t11=Confident Cloudy \n8\t Shadow Detected \n\t1=Yes 0=No \n9\t Cirrus Detection (IR)
(BTM15-BTM16) \n\t1=Cloud 0=No Cloud \n10 Snow/Ice \n 1=Snow/Ice, 0=No Snow/Ice \n";
        ubyte Snow_Flag(phony_dim_0, phony_dim_0);
            Snow_Flag: FillValue = 255UB;
            Snow_Flag:long_name = "Snow/Ice Status";
            Snow_Flag:units = "Unitless \n";
            Snow_Flag:valid_range = "0 - 1 \n";
            Snow_Flag:Description = "0 = No snow/ice \n 1 = snow/ice \n";
    } // group Data\Fields
} // group VNP_Grid_DNB
} // group GRIDS
} // group HDFEOS

group: HDFEOS\ INFORMATION {
    variables:
        string StructMetadata.0;

    // group attributes:
        :HDFEOSVersion = "HDFEOS_5.1.15";
} // group HDFEOS\ INFORMATION
}

```

## Appendix C: Metadata (Attributes) in VNP46A3 Product

Group size = 2

Number of attributes = 34

AlgorithmType = SCI

AlgorithmVersion = NPP\_PR46A3 2.0.0

Conventions = CF-1.6

DataResolution = 15 arc-second

DayNightFlag = Night

EndTime = 2018-08-01 00:00:00

HorizontalTileNumber = 06

InputPointer =

VNP46A1.A2018182.h06v05.001.2019194180841.h5,VNP46A1.A2018183.h06v05.001.2019194193646.h5,VNP46A1.A2018184.h06v05.001.2019194184433.h5,VNP46A1.A2018185.h06v05.001.2019194193216.h5,VNP46A1.A2018186.h06v05.001.2019194192910.h5,VNP46A1.A2018187.h06v05.001.2019194201314.h5,VNP46A1.A2018188.h06v05.001.2019194204007.h5,VNP46A1.A2018189.h06v05.001.2019194205610.h5,VNP46A1.A2018190.h06v05.001.2019194210836.h5,VNP46A1.A2018191.h06v05.001.2019194214016.h5,VNP46A1.A2018192.h06v05.001.2019194215926.h5,VNP46A1.A2018193.h06v05.001.2019194225222.h5,VNP46A1.A2018194.h06v05.001.2019194225425.h5,VNP46A1.A2018195.h06v05.001.2019194231815.h5,VNP46A1.A2018196.h06v05.001.2019194232434.h5,VNP46A1.A2018197.h06v05.001.2019194234549.h5,VNP46A1.A2018198.h06v05.001.2019195000500.h5,VNP46A1.A2018199.h06v05.001.2019195005010.h5,VNP46A1.A2018200.h06v05.001.2019195005953.h5,VNP46A1.A2018201.h06v05.001.2019195022017.h5,VNP46A1.A2018202.h06v05.001.2019195013145.h5,VNP46A1.A2018203.h06v05.001.2019195015640.h5,VNP46A1.A2018204.h06v05.001.2019195023109.h5,VNP46A1.A2018205.h06v05.001.2019196140441.h5,VNP46A1.A2018206.h06v05.001.2019196163318.h5,VNP46A1.A2018207.h06v05.001.2019196171208.h5,VNP46A1.A2018208.h06v05.001.2019196171354.h5,VNP46A1.A2018209.h06v05.001.2019196172238.h5,VNP46A1.A2018210.h06v05.001.2019196181758.h5,VNP46A1.A2018211.h06v05.001.2019196185014.h5,VNP46A1.A2018212.h06v05.001.2019196185126.h5,VNP46A2.A2018182.h06v05.001.2020343155445.h5,VNP46A2.A2018183.h06v05.001.2020343164259.h5,VNP46A2.A2018184.h06v05.001.2020343172948.h5,VNP46A2.A2018185.h06v05.001.2020343181842.h5,VNP46A2.A2018186.h06v05.001.2020343190314.h5,VNP46A2.A2018187.h06v05.001.2020343194858.h5,VNP46A2.A2018188.h06v05.001.2020343204138.h5,VNP46A2.A2018189.h06v05.001.2020343212856.h5,VNP46A2.A2018190.h06v05.001.2020343222642.h5,VNP46A2.A2018191.h06v05.001.2020343231735.h5,VNP46A2.A2018192.h06v05.001.2020343235424.h5,VNP46A2.A2018193.h06v05.001.2020344002902.h5,VNP46A2.A2018194.h06v05.001.2020344010341.h5,VNP46A2.A2018195.h06v05.001.2020344013818.h5,VNP46A2.A2018196.h06v05.001.2020344021310.h5,VNP46A2.A2018197.h06v05.001.2020344024631.h5,VNP46A2.A2018198.h06v05.001.2020344031936.h5,VNP46A2.A2018199.h06v05.001.2020344035615.h5,VNP46A2.A2018200.h06v05.001.2020344043242.h5,VNP46A2.A2018201.h06v05.001.2020344050922.h5,VNP46A2.A2018202.h06v05.001.2020344054229.h5,VNP46A2.A2018203.h06v05.001.2020344061554.h5,VNP46A2.A2018204.h06v05.001.2020344064733.h5,VNP46A2.A2018205.h06v05.001.2020344072948.h5,VNP46A2.A2018206.h06v05.001.2020344080837.h5,VNP46A2.A2018207.h06v05.001.2020344091604.h5,VNP46A2.A2018208.h06v05.001.2020344101454.h5,VNP46A2.A2018209.h06v05.001.2020344105623.h5,VNP46A2.A2018210.h06v05.001.2020344120527.h5,VNP46A2.A2018211.h06v05.001.2020344134430.h5,VNP46A2.A2018212.h06v05.001.2020344152335.h5

LocalGranuleID = VNP46A3.A2018182.h06v05.002.2021056211943.h5

LongName = VIIRS/NPP Lunar BRDF-Adjusted Nighttime Lights Monthly L3 Global 15 arc-second Linear Lat Lon Grid

NumberofInputGranules = 31

PGENumber = 556

PGEVersion = 2.0.0

```

PGE_EndTime = 2018-08-01 00:00:00.000
PGE_Name = PGE556
PGE_StartTime = 2018-07-01 00:00:00.000
PlatformShortName = SUOMI-NPP
ProcessVersion = 002
ProcessingCenter = LandSIPS
ProcessingEnvironment = Linux minion7092 3.10.0-1127.18.2.el7.x86_64 #1 SMP Sun Jul 26
15:27:06 UTC 2020 x86_64 x86_64 x86_64 GNU/Linux
ProductionTime = 2021-02-25 21:19:43.000
RangeBeginningDate = 2018-07-01
RangeBeginningTime = 00:00:00.000000
RangeEndingDate = 2018-07-31
RangeEndingTime = 23:59:00.000000
SatelliteInstrument = NPP_OPS
SensorShortname = VIIRS
ShortName = VNP46A3
StartTime = 2018-07-01 00:00:00
TileID = 61006005
VersionID = 002
VerticalTileNumber = 05
identifier_product_doi = 10.5067/VIIRS/VNP46A3.002
identifier_product_doi_authority = https://doi.org
variables:
    short AllAngle_Composite_Snow_Covered(fakeDim0, fakeDim1) ;
        AllAngle_Composite_Snow_Covered: FillValue = -1s ;
        AllAngle_Composite_Snow_Covered:long_name = "Temporal Radiance Composite Using All
Observations During Snow-covered Period" ;
        AllAngle_Composite_Snow_Covered:units = "nWatts/(cm^2 sr)" ;
        AllAngle_Composite_Snow_Covered:valid_range = 0s, -2s ;
        AllAngle_Composite_Snow_Covered:scale_factor = 0.1 ;
        AllAngle_Composite_Snow_Covered:offset = 0. ;
        AllAngle_Composite_Snow_Covered:coordinates = "latitude longitude" ;
    short AllAngle_Composite_Snow_Covered_Num(fakeDim2, fakeDim3) ;
        AllAngle_Composite_Snow_Covered_Num: FillValue = -1s ;
        AllAngle_Composite_Snow_Covered_Num:long_name = "Number of Observations of
Temporal Radiance Composite Using All Observations During Snow-covered Period" ;
        AllAngle_Composite_Snow_Covered_Num:units = "number of observations" ;
        AllAngle_Composite_Snow_Covered_Num:valid_range = 0s, -2s ;
        AllAngle_Composite_Snow_Covered_Num:scale_factor = 1. ;
        AllAngle_Composite_Snow_Covered_Num:offset = 0. ;
        AllAngle_Composite_Snow_Covered_Num:coordinates = "latitude longitude" ;
    byte AllAngle_Composite_Snow_Covered_Quality(fakeDim4, fakeDim5) ;
        AllAngle_Composite_Snow_Covered_Quality: FillValue = '\377' ;
        AllAngle_Composite_Snow_Covered_Quality:long_name = "Quality Flag of Temporal
Radiance Composite Using All Observations During Snow-covered Period" ;
        AllAngle_Composite_Snow_Covered_Quality:units = "quality flag, no units" ;
        AllAngle_Composite_Snow_Covered_Quality:valid_range = '\0', '\376' ;
        AllAngle_Composite_Snow_Covered_Quality:scale_factor = 1. ;
        AllAngle_Composite_Snow_Covered_Quality:offset = 0. ;
        AllAngle_Composite_Snow_Covered_Quality:Description = "Quality:\n",
"\t 0 = Good quality\n",

```

```

"\t 1 = Poor quality\n",
"\t 2 = Gap filled\n",
"\t 255 = Fill value\n",
"\t" ;
    AllAngle_Composite_Snow_Covered_Quality:coordinates = "latitude longitude" ;
short AllAngle_Composite_Snow_Covered_Std(fakeDim6, fakeDim7) ;
    AllAngle_Composite_Snow_Covered_Std: FillValue = -1s ;
    AllAngle_Composite_Snow_Covered_Std:long_name = "Standard Deviation of Temporal
Radiance Composite Using All Observations During Snow-covered Period" ;
    AllAngle_Composite_Snow_Covered_Std:units = "nWatts/(cm^2 sr)" ;
    AllAngle_Composite_Snow_Covered_Std:valid_range = 0s, -2s ;
    AllAngle_Composite_Snow_Covered_Std:scale_factor = 0.1 ;
    AllAngle_Composite_Snow_Covered_Std:offset = 0. ;
    AllAngle_Composite_Snow_Covered_Std:coordinates = "latitude longitude" ;
short AllAngle_Composite_Snow_Free(fakeDim8, fakeDim9) ;
    AllAngle_Composite_Snow_Free: FillValue = -1s ;
    AllAngle_Composite_Snow_Free:long_name = "Temporal Radiance Composite Using All
Observations During Snow-free Period" ;
    AllAngle_Composite_Snow_Free:units = "nWatts/(cm^2 sr)" ;
    AllAngle_Composite_Snow_Free:valid_range = 0s, -2s ;
    AllAngle_Composite_Snow_Free:scale_factor = 0.1 ;
    AllAngle_Composite_Snow_Free:offset = 0. ;
    AllAngle_Composite_Snow_Free:coordinates = "latitude longitude" ;
short AllAngle_Composite_Snow_Free_Num(fakeDim10, fakeDim11) ;
    AllAngle_Composite_Snow_Free_Num: FillValue = -1s ;
    AllAngle_Composite_Snow_Free_Num:long_name = "Number of Observations of Temporal
Radiance Composite Using All Observations During Snow-free Period" ;
    AllAngle_Composite_Snow_Free_Num:units = "number of observations" ;
    AllAngle_Composite_Snow_Free_Num:valid_range = 0s, -2s ;
    AllAngle_Composite_Snow_Free_Num:scale_factor = 1. ;
    AllAngle_Composite_Snow_Free_Num:offset = 0. ;
    AllAngle_Composite_Snow_Free_Num:coordinates = "latitude longitude" ;
byte AllAngle_Composite_Snow_Free_Quality(fakeDim12, fakeDim13) ;
    AllAngle_Composite_Snow_Free_Quality: FillValue = '\377' ;
    AllAngle_Composite_Snow_Free_Quality:long_name = "Quality Flag of Temporal Radiance
Composite Using All Observations During Snow-free Period" ;
    AllAngle_Composite_Snow_Free_Quality:units = "quality flag, no units" ;
    AllAngle_Composite_Snow_Free_Quality:valid_range = '\0', '\376' ;
    AllAngle_Composite_Snow_Free_Quality:scale_factor = 1. ;
    AllAngle_Composite_Snow_Free_Quality:offset = 0. ;
    AllAngle_Composite_Snow_Free_Quality:Description = "Quality:\n",
"\t 0 = Good quality\n",
"\t 1 = Poor quality\n",
"\t 2 = Gap filled\n",
"\t 255 = Fill value\n",
"\t" ;
    AllAngle_Composite_Snow_Free_Quality:coordinates = "latitude longitude" ;
short AllAngle_Composite_Snow_Free_Std(fakeDim14, fakeDim15) ;
    AllAngle_Composite_Snow_Free_Std: FillValue = -1s ;
    AllAngle_Composite_Snow_Free_Std:long_name = "Standard Deviation of Temporal
Radiance Composite Using All Observations During Snow-free Period" ;

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AllAngle_Composite_Snow_Free_Std:units = "nWatts/(cm^2 sr)" ;
AllAngle_Composite_Snow_Free_Std:valid_range = 0s, -2s ;
AllAngle_Composite_Snow_Free_Std:scale_factor = 0.1 ;
AllAngle_Composite_Snow_Free_Std:offset = 0. ;
AllAngle_Composite_Snow_Free_Std:coordinates = "latitude longitude" ;
byte DNB_Platform(fakeDim16, fakeDim17) ;
  DNB_Platform:_FillValue = '\377' ;
  DNB_Platform:long_name = "Platform" ;
  DNB_Platform:units = "platform, no units" ;
  DNB_Platform:valid_range = '\0', '\376' ;
  DNB_Platform:scale_factor = 1. ;
  DNB_Platform:offset = 0. ;
  DNB_Platform:Description = "Platform:\n",
"\t 0 = Suomi-NPP\n",
"\t 1 = NOAA-20\n",
"\t 2 = Suomi-NPP and NOAA-20 combined \n",
"\t" ;
  DNB_Platform:coordinates = "latitude longitude" ;
byte Land_Water_Mask(fakeDim18, fakeDim19) ;
  Land_Water_Mask:_FillValue = '\377' ;
  Land_Water_Mask:long_name = "Land Water Mask" ;
  Land_Water_Mask:units = "land water mask, no units" ;
  Land_Water_Mask:valid_range = '\0', '\376' ;
  Land_Water_Mask:scale_factor = 1. ;
  Land_Water_Mask:offset = 0. ;
  Land_Water_Mask:Description = "Land/Water:\n",
"\t 0 = Land & Desert \n",
"\t 1 = Land no Desert \n",
"\t 2 = Inland Water \n",
"\t 3 = Sea Water \n",
"\t 5 = Coastal \n",
"\t" ;
  Land_Water_Mask:coordinates = "latitude longitude" ;
short NearNadir_Composite_Snow_Covered(fakeDim20, fakeDim21) ;
  NearNadir_Composite_Snow_Covered:_FillValue = -1s ;
  NearNadir_Composite_Snow_Covered:long_name = "Temporal Radiance Composite Using
Nadir Angle Observations (View Zenith Angle 0-20 degree) During Snow-covered Period" ;
  NearNadir_Composite_Snow_Covered:units = "nWatts/(cm^2 sr)" ;
  NearNadir_Composite_Snow_Covered:valid_range = 0s, -2s ;
  NearNadir_Composite_Snow_Covered:scale_factor = 0.1 ;
  NearNadir_Composite_Snow_Covered:offset = 0. ;
  NearNadir_Composite_Snow_Covered:coordinates = "latitude longitude" ;
short NearNadir_Composite_Snow_Covered_Num(fakeDim22, fakeDim23) ;
  NearNadir_Composite_Snow_Covered_Num:_FillValue = -1s ;
  NearNadir_Composite_Snow_Covered_Num:long_name = "Number of Observations of
Temporal Radiance Composite Using Nadir Angle Observations (View Zenith Angle 0-20 degree) During
Snow-covered Period" ;
  NearNadir_Composite_Snow_Covered_Num:units = "number of observations" ;
  NearNadir_Composite_Snow_Covered_Num:valid_range = 0s, -2s ;
  NearNadir_Composite_Snow_Covered_Num:scale_factor = 1. ;
  NearNadir_Composite_Snow_Covered_Num:offset = 0. ;

```

```

NearNadir_Composite_Snow_Covered_Num:coordinates = "latitude longitude" ;
byte NearNadir_Composite_Snow_Covered_Quality(fakeDim24, fakeDim25) ;
NearNadir_Composite_Snow_Covered_Quality:_FillValue = '\377' ;
NearNadir_Composite_Snow_Covered_Quality:long_name = "Quality Flag of Temporal
Radiance Composite Using Nadir Angle Observations (View Zenith Angle 0-20 degree) During Snow-
covered Period" ;
NearNadir_Composite_Snow_Covered_Quality:units = "quality flag, no units" ;
NearNadir_Composite_Snow_Covered_Quality:valid_range = '\0', '\376' ;
NearNadir_Composite_Snow_Covered_Quality:scale_factor = 1. ;
NearNadir_Composite_Snow_Covered_Quality:offset = 0. ;
NearNadir_Composite_Snow_Covered_Quality:Description = "Quality:\n",
"\t 0 = Good quality\n",
"\t 1 = Poor quality\n",
"\t 2 = Gap filled\n",
"\t 255 = Fill value\n",
"\t" ;
NearNadir_Composite_Snow_Covered_Quality:coordinates = "latitude longitude" ;
short NearNadir_Composite_Snow_Covered_Std(fakeDim26, fakeDim27) ;
NearNadir_Composite_Snow_Covered_Std:_FillValue = -1s ;
NearNadir_Composite_Snow_Covered_Std:long_name = "Standard Deviation of Temporal
Radiance Composite Using Nadir Angle Observations (View Zenith Angle 0-20 degree) During Snow-
covered Period" ;
NearNadir_Composite_Snow_Covered_Std:units = "nWatts/(cm^2 sr)" ;
NearNadir_Composite_Snow_Covered_Std:valid_range = 0s, -2s ;
NearNadir_Composite_Snow_Covered_Std:scale_factor = 0.1 ;
NearNadir_Composite_Snow_Covered_Std:offset = 0. ;
NearNadir_Composite_Snow_Covered_Std:coordinates = "latitude longitude" ;
short NearNadir_Composite_Snow_Free(fakeDim28, fakeDim29) ;
NearNadir_Composite_Snow_Free:_FillValue = -1s ;
NearNadir_Composite_Snow_Free:long_name = "Temporal Radiance Composite Using Nadir
Angle Observations (View Zenith Angle 0-20 degree) During Snow-free Period" ;
NearNadir_Composite_Snow_Free:units = "nWatts/(cm^2 sr)" ;
NearNadir_Composite_Snow_Free:valid_range = 0s, -2s ;
NearNadir_Composite_Snow_Free:scale_factor = 0.1 ;
NearNadir_Composite_Snow_Free:offset = 0. ;
NearNadir_Composite_Snow_Free:coordinates = "latitude longitude" ;
short NearNadir_Composite_Snow_Free_Num(fakeDim30, fakeDim31) ;
NearNadir_Composite_Snow_Free_Num:_FillValue = -1s ;
NearNadir_Composite_Snow_Free_Num:long_name = "Number of Observations of Temporal
Radiance Composite Using Nadir Angle Observations (View Zenith Angle 0-20 degree) During Snow-
free Period" ;
NearNadir_Composite_Snow_Free_Num:units = "number of observations" ;
NearNadir_Composite_Snow_Free_Num:valid_range = 0s, -2s ;
NearNadir_Composite_Snow_Free_Num:scale_factor = 1. ;
NearNadir_Composite_Snow_Free_Num:offset = 0. ;
NearNadir_Composite_Snow_Free_Num:coordinates = "latitude longitude" ;
byte NearNadir_Composite_Snow_Free_Quality(fakeDim32, fakeDim33) ;
NearNadir_Composite_Snow_Free_Quality:_FillValue = '\377' ;
NearNadir_Composite_Snow_Free_Quality:long_name = "Quality Flag of Temporal Radiance
Composite Using Nadir Angle Observations (View Zenith Angle 0-20 degree) During Snow-free Period"
;

```

```

NearNadir_Composite_Snow_Free_Quality:units = "quality flag, no units" ;
NearNadir_Composite_Snow_Free_Quality:valid_range = '\0', '\376' ;
NearNadir_Composite_Snow_Free_Quality:scale_factor = 1. ;
NearNadir_Composite_Snow_Free_Quality:offset = 0. ;
NearNadir_Composite_Snow_Free_Quality:Description = "Quality:\n",
"\t 0 = Good quality\n",
"\t 1 = Poor quality\n",
"\t 2 = Gap filled\n",
"\t 255 = Fill value\n",
"\t" ;
NearNadir_Composite_Snow_Free_Quality:coordinates = "latitude longitude" ;
short NearNadir_Composite_Snow_Free_Std(fakeDim34, fakeDim35) ;
NearNadir_Composite_Snow_Free_Std:_FillValue = -1s ;
NearNadir_Composite_Snow_Free_Std:long_name = "Standard Deviation of Temporal
Radiance Composite Using Nadir Angle Observations (View Zenith Angle 0-20 degree) During Snow-
free Period" ;
NearNadir_Composite_Snow_Free_Std:units = "nWatts/(cm^2 sr)" ;
NearNadir_Composite_Snow_Free_Std:valid_range = 0s, -2s ;
NearNadir_Composite_Snow_Free_Std:scale_factor = 0.1 ;
NearNadir_Composite_Snow_Free_Std:offset = 0. ;
NearNadir_Composite_Snow_Free_Std:coordinates = "latitude longitude" ;
short OffNadir_Composite_Snow_Covered(fakeDim36, fakeDim37) ;
OffNadir_Composite_Snow_Covered:_FillValue = -1s ;
OffNadir_Composite_Snow_Covered:long_name = "Temporal Radiance Composite Using Off
Nadir Angle Observations (View Zenith Angle 40-60 degree) During Snow-covered Period" ;
OffNadir_Composite_Snow_Covered:units = "nWatts/(cm^2 sr)" ;
OffNadir_Composite_Snow_Covered:valid_range = 0s, -2s ;
OffNadir_Composite_Snow_Covered:scale_factor = 0.1 ;
OffNadir_Composite_Snow_Covered:offset = 0. ;
OffNadir_Composite_Snow_Covered:coordinates = "latitude longitude" ;
short OffNadir_Composite_Snow_Covered_Num(fakeDim38, fakeDim39) ;
OffNadir_Composite_Snow_Covered_Num:_FillValue = -1s ;
OffNadir_Composite_Snow_Covered_Num:long_name = "Number of Observations of
Temporal Radiance Composite Using Off Nadir Angle Observations (View Zenith Angle 40-60 degree)
During Snow-covered Period" ;
OffNadir_Composite_Snow_Covered_Num:units = "number of observations" ;
OffNadir_Composite_Snow_Covered_Num:valid_range = 0s, -2s ;
OffNadir_Composite_Snow_Covered_Num:scale_factor = 1. ;
OffNadir_Composite_Snow_Covered_Num:offset = 0. ;
OffNadir_Composite_Snow_Covered_Num:coordinates = "latitude longitude" ;
byte OffNadir_Composite_Snow_Covered_Quality(fakeDim40, fakeDim41) ;
OffNadir_Composite_Snow_Covered_Quality:_FillValue = '\377' ;
OffNadir_Composite_Snow_Covered_Quality:long_name = "Quality Flag of Temporal
Radiance Composite Using Off Nadir Angle Observations (View Zenith Angle 40-60 degree) During
Snow-covered Period" ;
OffNadir_Composite_Snow_Covered_Quality:units = "quality flag, no units" ;
OffNadir_Composite_Snow_Covered_Quality:valid_range = '\0', '\376' ;
OffNadir_Composite_Snow_Covered_Quality:scale_factor = 1. ;
OffNadir_Composite_Snow_Covered_Quality:offset = 0. ;
OffNadir_Composite_Snow_Covered_Quality:Description = "Quality:\n",
"\t 0 = Good quality\n",

```



```

"\t 1 = Poor quality\n",
"\t 2 = Gap filled\n",
"\t 255 = Fill value\n",
"\t" ;
    OffNadir_Composite_Snow_Covered_Quality:coordinates = "latitude longitude" ;
short OffNadir_Composite_Snow_Covered_Std(fakeDim42, fakeDim43) ;
    OffNadir_Composite_Snow_Covered_Std: FillValue = -1s ;
    OffNadir_Composite_Snow_Covered_Std:long_name = "Standard Deviation of Temporal
Radiance Composite Using Off Nadir Angle Observations (View Zenith Angle 40-60 degree) During
Snow-covered Period" ;
    OffNadir_Composite_Snow_Covered_Std:units = "nWatts/(cm^2 sr)" ;
    OffNadir_Composite_Snow_Covered_Std:valid_range = 0s, -2s ;
    OffNadir_Composite_Snow_Covered_Std:scale_factor = 0.1 ;
    OffNadir_Composite_Snow_Covered_Std:offset = 0. ;
    OffNadir_Composite_Snow_Covered_Std:coordinates = "latitude longitude" ;
short OffNadir_Composite_Snow_Free(fakeDim44, fakeDim45) ;
    OffNadir_Composite_Snow_Free: FillValue = -1s ;
    OffNadir_Composite_Snow_Free:units = "nWatts/(cm^2 sr)" ;
    OffNadir_Composite_Snow_Free:long_name = "Temporal Radiance Composite Using Off
Nadir Angle Observations (View Zenith Angle 40-60 degree) During Snow-free Period" ;
    OffNadir_Composite_Snow_Free:valid_range = 0s, -2s ;
    OffNadir_Composite_Snow_Free:scale_factor = 0.1 ;
    OffNadir_Composite_Snow_Free:offset = 0. ;
    OffNadir_Composite_Snow_Free:coordinates = "latitude longitude" ;
short OffNadir_Composite_Snow_Free_Num(fakeDim46, fakeDim47) ;
    OffNadir_Composite_Snow_Free_Num: FillValue = -1s ;
    OffNadir_Composite_Snow_Free_Num:long_name = "Number of Observations of Temporal
Radiance Composite Using Off Nadir Angle Observations (View Zenith Angle 40-60 degree) During
Snow-free Period" ;
    OffNadir_Composite_Snow_Free_Num:units = "number of observations" ;
    OffNadir_Composite_Snow_Free_Num:valid_range = 0s, -2s ;
    OffNadir_Composite_Snow_Free_Num:scale_factor = 1. ;
    OffNadir_Composite_Snow_Free_Num:offset = 0. ;
    OffNadir_Composite_Snow_Free_Num:coordinates = "latitude longitude" ;
byte OffNadir_Composite_Snow_Free_Quality(fakeDim48, fakeDim49) ;
    OffNadir_Composite_Snow_Free_Quality: FillValue = '\377' ;
    OffNadir_Composite_Snow_Free_Quality:long_name = "Quality Flag of Temporal Radiance
Composite Using Off Nadir Angle Observations (View Zenith Angle 40-60 degree) During Snow-free
Period" ;
    OffNadir_Composite_Snow_Free_Quality:units = "quality flag, no units" ;
    OffNadir_Composite_Snow_Free_Quality:valid_range = '\0', '\376' ;
    OffNadir_Composite_Snow_Free_Quality:scale_factor = 1. ;
    OffNadir_Composite_Snow_Free_Quality:offset = 0. ;
    OffNadir_Composite_Snow_Free_Quality:Description = "Quality:\n",
"\t 0 = Good quality\n",
"\t 1 = Poor quality\n",
"\t 2 = Gap filled\n",
"\t 255 = Fill value\n",
"\t" ;
    OffNadir_Composite_Snow_Free_Quality:coordinates = "latitude longitude" ;
short OffNadir_Composite_Snow_Free_Std(fakeDim50, fakeDim51) ;

```

```

OffNadir_Composite_Snow_Free_Std: FillValue = -1s ;
OffNadir_Composite_Snow_Free_Std:long_name = "Standard Deviation of Temporal Radiance
Composite Using Off Nadir Angle Observations (View Zenith Angle 40-60 degree) During Snow-free
Period" ;
OffNadir_Composite_Snow_Free_Std:units = "nWatts/(cm^2 sr)" ;
OffNadir_Composite_Snow_Free_Std:valid_range = 0s, -2s ;
OffNadir_Composite_Snow_Free_Std:scale_factor = 0.1 ;
OffNadir_Composite_Snow_Free_Std:offset = 0. ;
OffNadir_Composite_Snow_Free_Std:coordinates = "latitude longitude" ;
double lat(fakeDim52) ;
lat:long_name = "latitude" ;
lat:units = "degrees_north" ;
lat:_CoordinateAxisType = "Lat" ;
double lon(fakeDim53) ;
lon:long_name = "longitude" ;
lon:units = "degrees_east" ;
lon:_CoordinateAxisType = "Lon" ;

```

## Appendix D: Metadata (Attributes) in VNP46A4 Product

Group size = 2

Number of attributes = 34

AlgorithmType = SCI

AlgorithmVersion = NPP\_PR46A3 2.0.0

Conventions = CF-1.6

DataResolution = 15 arc-second

DayNightFlag = Night

EndTime = 2019-01-01 00:00:00

HorizontalTileNumber = 06

InputPointer =

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ProductionTime = 2021-02-25 21:41:06.000

RangeBeginningDate = 2018-01-01

RangeBeginningTime = 00:00:00.000000

RangeEndingDate = 2018-12-31

RangeEndingTime = 23:59:00.000000

SatelliteInstrument = NPP\_OPS

SensorShortname = VIIRS

ShortName = VNP46A4

StartTime = 2018-01-01 00:00:00

TileID = 61006005

VersionID = 002

VerticalTileNumber = 05

identifier\_product\_doi = 10.5067/VIIRS/VNP46A4.002

identifier\_product\_doi\_authority = https://doi.org

variables:

short AllAngle\_Composite\_Snow\_Covered(fakeDim0, fakeDim1) ;

AllAngle\_Composite\_Snow\_Covered: FillValue = -1s ;

AllAngle\_Composite\_Snow\_Covered:long\_name = "Temporal Radiance Composite Using All

Observations During Snow-covered Period" ;

AllAngle\_Composite\_Snow\_Covered:units = "nWatts/(cm^2 sr)" ;

AllAngle\_Composite\_Snow\_Covered:valid\_range = 0s, -2s ;

AllAngle\_Composite\_Snow\_Covered:scale\_factor = 0.1 ;

AllAngle\_Composite\_Snow\_Covered:offset = 0. ;

AllAngle\_Composite\_Snow\_Covered:coordinates = "latitude longitude" ;

short AllAngle\_Composite\_Snow\_Covered\_Num(fakeDim2, fakeDim3) ;

AllAngle\_Composite\_Snow\_Covered\_Num: FillValue = -1s ;

AllAngle\_Composite\_Snow\_Covered\_Num:long\_name = "Number of Observations of

Temporal Radiance Composite Using All Observations During Snow-covered Period" ;

AllAngle\_Composite\_Snow\_Covered\_Num:units = "number of observations" ;

AllAngle\_Composite\_Snow\_Covered\_Num:valid\_range = 0s, -2s ;

AllAngle\_Composite\_Snow\_Covered\_Num:scale\_factor = 1. ;

AllAngle\_Composite\_Snow\_Covered\_Num:offset = 0. ;

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        AllAngle_Composite_Snow_Covered_Num:coordinates = "latitude longitude" ;
byte AllAngle_Composite_Snow_Covered_Quality(fakeDim4, fakeDim5) ;
    AllAngle_Composite_Snow_Covered_Quality: FillValue = '\377' ;
    AllAngle_Composite_Snow_Covered_Quality:long_name = "Quality Flag of Temporal
Radiance Composite Using All Observations During Snow-covered Period" ;
    AllAngle_Composite_Snow_Covered_Quality:units = "quality flag, no units" ;
    AllAngle_Composite_Snow_Covered_Quality:valid_range = '\0', '\376' ;
    AllAngle_Composite_Snow_Covered_Quality:scale_factor = 1. ;
    AllAngle_Composite_Snow_Covered_Quality:offset = 0. ;
    AllAngle_Composite_Snow_Covered_Quality:Description = "Quality:\n",
"\t 0 = Good quality\n",
"\t 1 = Poor quality\n",
"\t 2 = Gap filled\n",
"\t 255 = Fill value\n",
"\t" ;
        AllAngle_Composite_Snow_Covered_Quality:coordinates = "latitude longitude" ;
short AllAngle_Composite_Snow_Covered_Std(fakeDim6, fakeDim7) ;
    AllAngle_Composite_Snow_Covered_Std: FillValue = -1s ;
    AllAngle_Composite_Snow_Covered_Std:long_name = "Standard Deviation of Temporal
Radiance Composite Using All Observations During Snow-covered Period" ;
    AllAngle_Composite_Snow_Covered_Std:units = "nWatts/(cm^2 sr)" ;
    AllAngle_Composite_Snow_Covered_Std:valid_range = 0s, -2s ;
    AllAngle_Composite_Snow_Covered_Std:scale_factor = 0.1 ;
    AllAngle_Composite_Snow_Covered_Std:offset = 0. ;
    AllAngle_Composite_Snow_Covered_Std:coordinates = "latitude longitude" ;
short AllAngle_Composite_Snow_Free(fakeDim8, fakeDim9) ;
    AllAngle_Composite_Snow_Free: FillValue = -1s ;
    AllAngle_Composite_Snow_Free:long_name = "Temporal Radiance Composite Using All
Observations During Snow-free Period" ;
    AllAngle_Composite_Snow_Free:units = "nWatts/(cm^2 sr)" ;
    AllAngle_Composite_Snow_Free:valid_range = 0s, -2s ;
    AllAngle_Composite_Snow_Free:scale_factor = 0.1 ;
    AllAngle_Composite_Snow_Free:offset = 0. ;
    AllAngle_Composite_Snow_Free:coordinates = "latitude longitude" ;
short AllAngle_Composite_Snow_Free_Num(fakeDim10, fakeDim11) ;
    AllAngle_Composite_Snow_Free_Num: FillValue = -1s ;
    AllAngle_Composite_Snow_Free_Num:long_name = "Number of Observations of Temporal
Radiance Composite Using All Observations During Snow-free Period" ;
    AllAngle_Composite_Snow_Free_Num:units = "number of observations" ;
    AllAngle_Composite_Snow_Free_Num:valid_range = 0s, -2s ;
    AllAngle_Composite_Snow_Free_Num:scale_factor = 1. ;
    AllAngle_Composite_Snow_Free_Num:offset = 0. ;
    AllAngle_Composite_Snow_Free_Num:coordinates = "latitude longitude" ;
byte AllAngle_Composite_Snow_Free_Quality(fakeDim12, fakeDim13) ;
    AllAngle_Composite_Snow_Free_Quality: FillValue = '\377' ;
    AllAngle_Composite_Snow_Free_Quality:long_name = "Quality Flag of Temporal Radiance
Composite Using All Observations During Snow-free Period" ;
    AllAngle_Composite_Snow_Free_Quality:units = "quality flag, no units" ;
    AllAngle_Composite_Snow_Free_Quality:valid_range = '\0', '\376' ;
    AllAngle_Composite_Snow_Free_Quality:scale_factor = 1. ;
    AllAngle_Composite_Snow_Free_Quality:offset = 0. ;

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        AllAngle_Composite_Snow_Free_Quality:Description = "Quality:\n",
"\t 0 = Good quality\n",
"\t 1 = Poor quality\n",
"\t 2 = Gap filled\n",
"\t 255 = Fill value\n",
"\t" ;
        AllAngle_Composite_Snow_Free_Quality:coordinates = "latitude longitude" ;
short AllAngle_Composite_Snow_Free_Std(fakeDim14, fakeDim15) ;
        AllAngle_Composite_Snow_Free_Std: FillValue = -1s ;
        AllAngle_Composite_Snow_Free_Std:long_name = "Standard Deviation of Temporal
Radiance Composite Using All Observations During Snow-free Period" ;
        AllAngle_Composite_Snow_Free_Std:units = "nWatts/(cm^2 sr)" ;
        AllAngle_Composite_Snow_Free_Std:valid_range = 0s, -2s ;
        AllAngle_Composite_Snow_Free_Std:scale_factor = 0.1 ;
        AllAngle_Composite_Snow_Free_Std:offset = 0. ;
        AllAngle_Composite_Snow_Free_Std:coordinates = "latitude longitude" ;
byte DNB_Platform(fakeDim16, fakeDim17) ;
        DNB_Platform: FillValue = '\377' ;
        DNB_Platform:long_name = "Platform" ;
        DNB_Platform:units = "platform, no units" ;
        DNB_Platform:valid_range = '\0', '\376' ;
        DNB_Platform:scale_factor = 1. ;
        DNB_Platform:offset = 0. ;
        DNB_Platform:Description = "Platform:\n",
"\t 0 = Suomi-NPP\n",
"\t 1 = NOAA-20\n",
"\t 2 = Suomi-NPP and NOAA-20 combined \n",
"\t" ;
        DNB_Platform:coordinates = "latitude longitude" ;
byte Land_Water_Mask(fakeDim18, fakeDim19) ;
        Land_Water_Mask: FillValue = '\377' ;
        Land_Water_Mask:long_name = "Land Water Mask" ;
        Land_Water_Mask:units = "land water mask, no units" ;
        Land_Water_Mask:valid_range = '\0', '\376' ;
        Land_Water_Mask:scale_factor = 1. ;
        Land_Water_Mask:offset = 0. ;
        Land_Water_Mask:Description = "Land/Water:\n",
"\t 0 = Land & Desert \n",
"\t 1 = Land no Desert \n",
"\t 2 = Inland Water \n",
"\t 3 = Sea Water \n",
"\t 5 = Coastal \n",
"\t" ;
        Land_Water_Mask:coordinates = "latitude longitude" ;
short NearNadir_Composite_Snow_Covered(fakeDim20, fakeDim21) ;
        NearNadir_Composite_Snow_Covered: FillValue = -1s ;
        NearNadir_Composite_Snow_Covered:long_name = "Temporal Radiance Composite Using
Nadir Angle Observations (View Zenith Angle 0-20 degree) During Snow-covered Period" ;
        NearNadir_Composite_Snow_Covered:units = "nWatts/(cm^2 sr)" ;
        NearNadir_Composite_Snow_Covered:valid_range = 0s, -2s ;
        NearNadir_Composite_Snow_Covered:scale_factor = 0.1 ;

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NearNadir_Composite_Snow_Covered:offset = 0. ;
NearNadir_Composite_Snow_Covered:coordinates = "latitude longitude" ;
short NearNadir_Composite_Snow_Covered_Num(fakeDim22, fakeDim23) ;
NearNadir_Composite_Snow_Covered_Num:_FillValue = -1s ;
NearNadir_Composite_Snow_Covered_Num:long_name = "Number of Observations of
Temporal Radiance Composite Using Nadir Angle Observations (View Zenith Angle 0-20 degree) During
Snow-covered Period" ;
NearNadir_Composite_Snow_Covered_Num:units = "number of observations" ;
NearNadir_Composite_Snow_Covered_Num:valid_range = 0s, -2s ;
NearNadir_Composite_Snow_Covered_Num:scale_factor = 1. ;
NearNadir_Composite_Snow_Covered_Num:offset = 0. ;
NearNadir_Composite_Snow_Covered_Num:coordinates = "latitude longitude" ;
byte NearNadir_Composite_Snow_Covered_Quality(fakeDim24, fakeDim25) ;
NearNadir_Composite_Snow_Covered_Quality:_FillValue = '\377' ;
NearNadir_Composite_Snow_Covered_Quality:long_name = "Quality Flag of Temporal
Radiance Composite Using Nadir Angle Observations (View Zenith Angle 0-20 degree) During Snow-
covered Period" ;
NearNadir_Composite_Snow_Covered_Quality:units = "quality flag, no units" ;
NearNadir_Composite_Snow_Covered_Quality:valid_range = '\0', '\376' ;
NearNadir_Composite_Snow_Covered_Quality:scale_factor = 1. ;
NearNadir_Composite_Snow_Covered_Quality:offset = 0. ;
NearNadir_Composite_Snow_Covered_Quality:Description = "Quality:\n",
"\t 0 = Good quality\n",
"\t 1 = Poor quality\n",
"\t 2 = Gap filled\n",
"\t 255 = Fill value\n",
"\t" ;
NearNadir_Composite_Snow_Covered_Quality:coordinates = "latitude longitude" ;
short NearNadir_Composite_Snow_Covered_Std(fakeDim26, fakeDim27) ;
NearNadir_Composite_Snow_Covered_Std:_FillValue = -1s ;
NearNadir_Composite_Snow_Covered_Std:long_name = "Standard Deviation of Temporal
Radiance Composite Using Nadir Angle Observations (View Zenith Angle 0-20 degree) During Snow-
covered Period" ;
NearNadir_Composite_Snow_Covered_Std:units = "nWatts/(cm^2 sr)" ;
NearNadir_Composite_Snow_Covered_Std:valid_range = 0s, -2s ;
NearNadir_Composite_Snow_Covered_Std:scale_factor = 0.1 ;
NearNadir_Composite_Snow_Covered_Std:offset = 0. ;
NearNadir_Composite_Snow_Covered_Std:coordinates = "latitude longitude" ;
short NearNadir_Composite_Snow_Free(fakeDim28, fakeDim29) ;
NearNadir_Composite_Snow_Free:_FillValue = -1s ;
NearNadir_Composite_Snow_Free:long_name = "Temporal Radiance Composite Using Nadir
Angle Observations (View Zenith Angle 0-20 degree) During Snow-free Period" ;
NearNadir_Composite_Snow_Free:units = "nWatts/(cm^2 sr)" ;
NearNadir_Composite_Snow_Free:valid_range = 0s, -2s ;
NearNadir_Composite_Snow_Free:scale_factor = 0.1 ;
NearNadir_Composite_Snow_Free:offset = 0. ;
NearNadir_Composite_Snow_Free:coordinates = "latitude longitude" ;
short NearNadir_Composite_Snow_Free_Num(fakeDim30, fakeDim31) ;
NearNadir_Composite_Snow_Free_Num:_FillValue = -1s ;

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NearNadir_Composite_Snow_Free_Num:long_name = "Number of Observations of Temporal
Radiance Composite Using Nadir Angle Observations (View Zenith Angle 0-20 degree) During Snow-
free Period" ;
NearNadir_Composite_Snow_Free_Num:units = "number of observations" ;
NearNadir_Composite_Snow_Free_Num:valid_range = 0s, -2s ;
NearNadir_Composite_Snow_Free_Num:scale_factor = 1. ;
NearNadir_Composite_Snow_Free_Num:offset = 0. ;
NearNadir_Composite_Snow_Free_Num:coordinates = "latitude longitude" ;
byte NearNadir_Composite_Snow_Free_Quality(fakeDim32, fakeDim33) ;
NearNadir_Composite_Snow_Free_Quality:_FillValue = '\377' ;
NearNadir_Composite_Snow_Free_Quality:long_name = "Quality Flag of Temporal Radiance
Composite Using Nadir Angle Observations (View Zenith Angle 0-20 degree) During Snow-free Period"
;
NearNadir_Composite_Snow_Free_Quality:units = "quality flag, no units" ;
NearNadir_Composite_Snow_Free_Quality:valid_range = '\0', '\376' ;
NearNadir_Composite_Snow_Free_Quality:scale_factor = 1. ;
NearNadir_Composite_Snow_Free_Quality:offset = 0. ;
NearNadir_Composite_Snow_Free_Quality:Description = "Quality:\n",
"\t 0 = Good quality\n",
"\t 1 = Poor quality\n",
"\t 2 = Gap filled\n",
"\t 255 = Fill value\n",
"\t" ;
NearNadir_Composite_Snow_Free_Quality:coordinates = "latitude longitude" ;
short NearNadir_Composite_Snow_Free_Std(fakeDim34, fakeDim35) ;
NearNadir_Composite_Snow_Free_Std:_FillValue = -1s ;
NearNadir_Composite_Snow_Free_Std:long_name = "Standard Deviation of Temporal
Radiance Composite Using Nadir Angle Observations (View Zenith Angle 0-20 degree) During Snow-
free Period" ;
NearNadir_Composite_Snow_Free_Std:units = "nWatts/(cm^2 sr)" ;
NearNadir_Composite_Snow_Free_Std:valid_range = 0s, -2s ;
NearNadir_Composite_Snow_Free_Std:scale_factor = 0.1 ;
NearNadir_Composite_Snow_Free_Std:offset = 0. ;
NearNadir_Composite_Snow_Free_Std:coordinates = "latitude longitude" ;
short OffNadir_Composite_Snow_Covered(fakeDim36, fakeDim37) ;
OffNadir_Composite_Snow_Covered:_FillValue = -1s ;
OffNadir_Composite_Snow_Covered:long_name = "Temporal Radiance Composite Using Off
Nadir Angle Observations (View Zenith Angle 40-60 degree) During Snow-covered Period" ;
OffNadir_Composite_Snow_Covered:units = "nWatts/(cm^2 sr)" ;
OffNadir_Composite_Snow_Covered:valid_range = 0s, -2s ;
OffNadir_Composite_Snow_Covered:scale_factor = 0.1 ;
OffNadir_Composite_Snow_Covered:offset = 0. ;
OffNadir_Composite_Snow_Covered:coordinates = "latitude longitude" ;
short OffNadir_Composite_Snow_Covered_Num(fakeDim38, fakeDim39) ;
OffNadir_Composite_Snow_Covered_Num:_FillValue = -1s ;
OffNadir_Composite_Snow_Covered_Num:long_name = "Number of Observations of
Temporal Radiance Composite Using Off Nadir Angle Observations (View Zenith Angle 40-60 degree)
During Snow-covered Period" ;
OffNadir_Composite_Snow_Covered_Num:units = "number of observations" ;
OffNadir_Composite_Snow_Covered_Num:valid_range = 0s, -2s ;
OffNadir_Composite_Snow_Covered_Num:scale_factor = 1. ;

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    OffNadir_Composite_Snow_Covered_Num:offset = 0. ;
    OffNadir_Composite_Snow_Covered_Num:coordinates = "latitude longitude" ;
byte OffNadir_Composite_Snow_Covered_Quality(fakeDim40, fakeDim41) ;
    OffNadir_Composite_Snow_Covered_Quality:_FillValue = '\377' ;
    OffNadir_Composite_Snow_Covered_Quality:long_name = "Quality Flag of Temporal
Radiance Composite Using Off Nadir Angle Observations (View Zenith Angle 40-60 degree) During
Snow-covered Period" ;
    OffNadir_Composite_Snow_Covered_Quality:units = "quality flag, no units" ;
    OffNadir_Composite_Snow_Covered_Quality:valid_range = '\0', '\376' ;
    OffNadir_Composite_Snow_Covered_Quality:scale_factor = 1. ;
    OffNadir_Composite_Snow_Covered_Quality:offset = 0. ;
    OffNadir_Composite_Snow_Covered_Quality:Description = "Quality:\n",
"\t 0 = Good quality\n",
"\t 1 = Poor quality\n",
"\t 2 = Gap filled\n",
"\t 255 = Fill value\n",
"\t" ;
    OffNadir_Composite_Snow_Covered_Quality:coordinates = "latitude longitude" ;
short OffNadir_Composite_Snow_Covered_Std(fakeDim42, fakeDim43) ;
    OffNadir_Composite_Snow_Covered_Std:_FillValue = -1s ;
    OffNadir_Composite_Snow_Covered_Std:long_name = "Standard Deviation of Temporal
Radiance Composite Using Off Nadir Angle Observations (View Zenith Angle 40-60 degree) During
Snow-covered Period" ;
    OffNadir_Composite_Snow_Covered_Std:units = "nWatts/(cm^2 sr)" ;
    OffNadir_Composite_Snow_Covered_Std:valid_range = 0s, -2s ;
    OffNadir_Composite_Snow_Covered_Std:scale_factor = 0.1 ;
    OffNadir_Composite_Snow_Covered_Std:offset = 0. ;
    OffNadir_Composite_Snow_Covered_Std:coordinates = "latitude longitude" ;
short OffNadir_Composite_Snow_Free(fakeDim44, fakeDim45) ;
    OffNadir_Composite_Snow_Free:_FillValue = -1s ;
    OffNadir_Composite_Snow_Free:units = "nWatts/(cm^2 sr)" ;
    OffNadir_Composite_Snow_Free:long_name = "Temporal Radiance Composite Using Off
Nadir Angle Observations (View Zenith Angle 40-60 degree) During Snow-free Period" ;
    OffNadir_Composite_Snow_Free:valid_range = 0s, -2s ;
    OffNadir_Composite_Snow_Free:scale_factor = 0.1 ;
    OffNadir_Composite_Snow_Free:offset = 0. ;
    OffNadir_Composite_Snow_Free:coordinates = "latitude longitude" ;
short OffNadir_Composite_Snow_Free_Num(fakeDim46, fakeDim47) ;
    OffNadir_Composite_Snow_Free_Num:_FillValue = -1s ;
    OffNadir_Composite_Snow_Free_Num:long_name = "Number of Observations of Temporal
Radiance Composite Using Off Nadir Angle Observations (View Zenith Angle 40-60 degree) During
Snow-free Period" ;
    OffNadir_Composite_Snow_Free_Num:units = "number of observations" ;
    OffNadir_Composite_Snow_Free_Num:valid_range = 0s, -2s ;
    OffNadir_Composite_Snow_Free_Num:scale_factor = 1. ;
    OffNadir_Composite_Snow_Free_Num:offset = 0. ;
    OffNadir_Composite_Snow_Free_Num:coordinates = "latitude longitude" ;
byte OffNadir_Composite_Snow_Free_Quality(fakeDim48, fakeDim49) ;
    OffNadir_Composite_Snow_Free_Quality:_FillValue = '\377' ;

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OffNadir_Composite_Snow_Free_Quality:long_name = "Quality Flag of Temporal Radiance
Composite Using Off Nadir Angle Observations (View Zenith Angle 40-60 degree) During Snow-free
Period" ;
OffNadir_Composite_Snow_Free_Quality:units = "quality flag, no units" ;
OffNadir_Composite_Snow_Free_Quality:valid_range = '0', '376' ;
OffNadir_Composite_Snow_Free_Quality:scale_factor = 1. ;
OffNadir_Composite_Snow_Free_Quality:offset = 0. ;
OffNadir_Composite_Snow_Free_Quality:Description = "Quality:\n",
"\t 0 = Good quality\n",
"\t 1 = Poor quality\n",
"\t 2 = Gap filled\n",
"\t 255 = Fill value\n",
"\t" ;
OffNadir_Composite_Snow_Free_Quality:coordinates = "latitude longitude" ;
short OffNadir_Composite_Snow_Free_Std(fakeDim50, fakeDim51) ;
OffNadir_Composite_Snow_Free_Std:_FillValue = -1s ;
OffNadir_Composite_Snow_Free_Std:long_name = "Standard Deviation of Temporal Radiance
Composite Using Off Nadir Angle Observations (View Zenith Angle 40-60 degree) During Snow-free
Period" ;
OffNadir_Composite_Snow_Free_Std:units = "nWatts/(cm^2 sr)" ;
OffNadir_Composite_Snow_Free_Std:valid_range = 0s, -2s ;
OffNadir_Composite_Snow_Free_Std:scale_factor = 0.1 ;
OffNadir_Composite_Snow_Free_Std:offset = 0. ;
OffNadir_Composite_Snow_Free_Std:coordinates = "latitude longitude" ;
double lat(fakeDim52) ;
lat:long_name = "latitude" ;
lat:units = "degrees_north" ;
lat:_CoordinateAxisType = "Lat" ;
double lon(fakeDim53) ;
lon:long_name = "longitude" ;
lon:units = "degrees_east" ;
lon:_CoordinateAxisType = "Lon" ;

```